

SUMS OF TWO SQUARES IN THREE-TERM PATTERNS

ABSTRACT. For fixed integers $0 < a < b$ let $S_{a,b}(x)$ be the number of $n \leq x$ such that $n, n+a$ and $n+b$ are all sums of two squares. We prove that $S_{a,b}(x) \gg x^{1/2+\delta}$ for every $\delta < (1-2\theta)/(10-12\theta)$, where θ is any exponent towards Selberg's eigenvalue conjecture for congruence subgroups. The Kim–Sarnak bound $\theta = 7/64$, that is, $\lambda_1 \geq 975/4096$ for the smallest positive Laplace eigenvalue, gives every $\delta < 25/278 = 0.0899\dots$, and Selberg's conjecture would give every $\delta < 1/10$.

Two of the three numbers are made sums of two squares by an identity. The third runs through the values $2^{v-2}(E\ell^2 + H_d)/d$ of a quadratic polynomial attached to a prime d , where $E \in \{4, 8\}$, $H_d \asymp d^2$, and all 2-adic conditions are imposed on d . The r -weighted count of these values leads to Type I sums for the roots of $E\ell^2 + H_d \equiv 0$ modulo dk with $k \equiv \pm 1 \pmod{4}$. We estimate them by the method of Duke, Friedlander and Iwaniec. Poisson summation turns them into Weyl sums, and these are Poincaré series on $\Gamma_0(4Ed)$ evaluated at Heegner points of discriminant $-4EH_d$. In the spectral expansion, the Fourier coefficients are controlled by large sieve inequalities that come from the Kuznetsov formula: those of Deshouillers and Iwaniec, and Pascadi's inequality for exceptional Maass forms and exponential phases, which we extend to smooth weights. The values at the Heegner points are controlled by the pre-trace formula and an elementary count of pairs of Heegner points at bounded distance. This admits primes d up to $x^{(1-2\theta)/(5-6\theta)-\varepsilon}$. With only the inequalities of Deshouillers and Iwaniec, the same argument gives every $\delta < (1-2\theta)/(10-8\theta)$.

1. INTRODUCTION

Let $\mathcal{S} = \{x^2 + y^2 : x, y \in \mathbb{Z}\}$. For fixed integers $0 < a < b$ put

$$S_{a,b}(x) = \#\{1 \leq n \leq x : n, n+a, n+b \in \mathcal{S}\}.$$

Littlewood asked whether $S_{a,b}(x) \rightarrow \infty$ for all such a, b . Hooley [15] answered this affirmatively, using the theory of ternary quadratic forms; see also [8, §2.1].

The expected order of magnitude comes from an analogue, for sums of two squares, of the Hardy–Littlewood prime k -tuple conjecture. It was formulated by Freiberg, Kurlberg and Rosenzweig [8, Conjecture 1.1], following Connors and Keating [2] and Smilansky [27] for pairs. For a finite set $\mathbf{h} \subset \mathbb{Z}$ with non-vanishing singular series $\mathfrak{S}_{\mathbf{h}}$ it predicts

$$\#\{n \leq x : n + \mathbf{h} \subseteq \mathcal{S}\} \sim \mathfrak{S}_{\mathbf{h}} x \left(\frac{C}{\sqrt{\log x}} \right)^{\#\mathbf{h}},$$

where $C = 0.7642\dots$ is the Landau–Ramanujan constant. Every three-element set has $\mathfrak{S}_{\mathbf{h}} > 0$ [8, §2.1], so the conjecture predicts $S_{a,b}(x) \sim \mathfrak{S}_{\{0,a,b\}} C^3 x (\log x)^{-3/2}$.

Upper bounds of this order follow from Selberg's sieve; this was done by Cochrane and Dressler [1] for $\{0, 1, 2\}$ and by Nowak [22] in general. Lower bounds are much weaker. For pairs, the bound $\#\{n \leq x : n, n+j \in \mathcal{S}\} \gg x/\log x$ of the conjectured order is due to Hooley [16] and Indlekofer [17], but, as noted in [8], no such bound is known for any triple. One-parameter constructions give at most about $x^{1/2}$ elements up to x . For example, for $\{0, 1, 2\}$, taking $n = m^2$ requires only $m^2 + 2 \in \mathcal{S}$, and [9, Theorem 2] gives $\gg X(\log X)^{-1/2}$ such $m \leq X$. As far as we are aware, no lower bound $S_{a,b}(x) \gg x^{1/2+\delta}$ with $\delta > 0$ was known for any pattern before the present work (see §1.3 for its earlier versions).

For a congruence subgroup $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$, let $\lambda_1(\Gamma)$ be the smallest positive eigenvalue of the Laplacian on $\Gamma \backslash \mathbb{H}$, and put $\theta(\Gamma) = \max\{0, \mathrm{Re} \sqrt{1/4 - \lambda_1(\Gamma)}\}$. We call $\theta \in [0, \frac{1}{2})$ *admissible* if $\theta(\Gamma_0(q)) \leq \theta$ for every $q \geq 1$. Since $\lambda_1(\Gamma) = \frac{1}{4} - \theta(\Gamma)^2$ whenever $\theta(\Gamma) > 0$, the admissibility

of θ means that $\lambda_1(\Gamma_0(q)) \geq \frac{1}{4} - \theta^2$ for every q . Selberg proved that $\lambda_1 \geq 3/16$, so $\theta = 1/4$ is admissible [24]. Kim and Sarnak proved that $\lambda_1(\Gamma) \geq 975/4096 = \frac{1}{4} - (\frac{7}{64})^2$ for every congruence subgroup Γ , so $\theta = 7/64$ is admissible [19, Appendix 2]; this is the strongest bound currently known. Selberg's eigenvalue conjecture states that $\lambda_1 \geq 1/4$, that is, that $\theta = 0$ is admissible.

Theorem 1.1. *Let $0 < a < b$ be fixed integers and let θ be admissible. For every*

$$\delta < \delta_\theta := \frac{1 - 2\theta}{10 - 12\theta}$$

we have $S_{a,b}(x) \gg_{a,b,\delta} x^{1/2+\delta}$ for $x \geq x_0(a, b, \delta)$.

Corollary 1.2. *For every $\delta < 25/278 = 0.0899\dots$ we have $S_{a,b}(x) \gg_{a,b,\delta} x^{1/2+\delta}$ for $x \geq x_0(a, b, \delta)$. If Selberg's eigenvalue conjecture holds, then this is true for every $\delta < 1/10$. In particular, for every $\delta < 25/278$ and all sufficiently large x there are $\gg_\delta x^{1/2+\delta}$ integers $n \leq x$ such that $n-1$, n and $n+1$ are all sums of two squares.*

The corollary is the case $\theta = 7/64$, respectively $\theta = 0$, of Theorem 1.1. The implied constants are ineffective because Siegel's theorem is used; Remark 13.6 explains how to avoid this.

The proof of Theorem 1.1 uses the large sieve inequality of Deshouillers and Iwaniec for the regular spectrum [4, Theorem 2] and Pascadi's large sieve inequality for exceptional Maass forms and exponential phases [23, Theorem 2]. If Pascadi's inequality is replaced by the large sieve inequality of Deshouillers and Iwaniec for the exceptional spectrum [4, Theorem 5], the same argument gives the following.

Theorem 1.3. *Let $0 < a < b$ be fixed integers and let θ be admissible. For every $\delta < \delta'_\theta := (1 - 2\theta)/(10 - 8\theta)$ we have $S_{a,b}(x) \gg_{a,b,\delta} x^{1/2+\delta}$ for $x \geq x_0(a, b, \delta)$.*

Remark 1.4. Besides elementary arguments, the proof of Theorem 1.3 uses Siegel's theorem (or the Landau–Page theorem, see Remark 13.6), the Pólya–Vinogradov inequality, the prime number theorem in arithmetic progressions, the spectral theory of $\Gamma_0(q) \backslash \mathbb{H}$ [18], classical facts about Bessel functions [29, 11], [4, Theorems 2 and 5], and the admissibility of θ . The proof of Theorem 1.1 uses [23, Theorem 2] in place of [4, Theorem 5].

Table 1 lists the resulting exponents.

θ	Theorem 1.1	Theorem 1.3
1/4 (Selberg)	1/14 = 0.0714	1/16 = 0.0625
7/64 (Kim–Sarnak)	25/278 = 0.0899	25/292 = 0.0856
0 (Selberg's conjecture)	1/10	1/10

TABLE 1. Exponents in $S_{a,b}(x) \gg x^{1/2+\delta}$: the bound holds for every δ below the value shown. The three values of θ correspond to $\lambda_1 \geq 3/16$, $\lambda_1 \geq 975/4096$ and $\lambda_1 \geq 1/4$.

1.1. The method. Two of the three numbers are made sums of two squares automatically. Let B be odd and put $Q = 2P + B$. If

$$2dQ = u^2 + d^2 + B^2$$

for integers d, u , then $4P(P + B) = (Q - d)^2 + u^2$. If d is moreover a prime with $d \equiv 1 \pmod{4}$, then P and $P + B$ are both sums of two squares (Lemma 2.5). The third number $T = P + A$ then satisfies

$$4dT = u^2 + m_d, \quad m_d = d^2 + 2cd + B^2, \quad c = 2A - B.$$

For a fixed prime d and u in a lattice, T runs through the values of a quadratic polynomial, and $\asymp \sqrt{x/d}$ of these are at most x . We show that the r -weighted count of these values has the expected size. Summing over primes $d \leq x^\varpi$ and removing multiplicities at a cost of x^ε then gives $S_{a,b}(x) \gg x^{1/2+\varpi/2-\varepsilon}$. Every pattern can be brought into this form, by choosing which of $n, n+a, n+b$ plays the role of T (Lemma 2.3). For the pattern $\{0, 1, 2\}$ the identity is closely related to an observation of Skalba [26].

The weighted count is evaluated by the hyperbola method, which requires T to be 2^{v-2} times an integer $\equiv 1 \pmod{4}$ for all u in a lattice. We impose all 2-adic conditions on d alone. By Proposition 3.2 there are $v \geq 2$ and d_0 with the following property: for every prime $d \equiv d_0 \pmod{2^{v+2}}$ and every $u \equiv 0 \pmod{2^j}$ with $d \mid u^2 + B^2$, where $j = \lceil (v+2)/2 \rceil$, the number T is 2^{v-2} times an integer $\equiv 1 \pmod{4}$. Writing $u = 2^j \ell$, the polynomial becomes $2^v(E\ell^2 + H_d)$ with $E \in \{4, 8\}$ and $H_d = 2^{-v}m_d \asymp d^2$. The hyperbola method then leads to Type I sums

$$\sum_{k \equiv \kappa \pmod{4}} \#\{\ell : dk \mid E\ell^2 + H_d, \dots\} \quad (\kappa \in \{1, 3\})$$

for the roots of a quadratic congruence. The moduli $\mu = dk$ go up to about $d\sqrt{x}$, which exceeds the length $\asymp \sqrt{dx}$ of the ℓ -sum by a factor $\asymp \sqrt{d}$. The character $\chi(k)$ of the hyperbola method has become the congruence condition $\mu \equiv \kappa d \pmod{4d}$.

Following Duke, Friedlander and Iwaniec [6], the roots of $E\ell^2 + H \equiv 0 \pmod{\mu}$, for all moduli μ at once, correspond to the integral binary quadratic forms of determinant $N = EH$, that is, to Heegner points of discriminant $-4N$. The condition $\mu \equiv \kappa d \pmod{4d}$ becomes a condition on symmetric matrices that is invariant under $\Gamma_0(q)$, $q = 4Ed$ (Lemma 8.2). Our main analytic result, Theorem 6.1, is a Type I estimate with square-root cancellation in the length X of the ℓ -sum, uniformly for moduli up to qX . The price is a factor $H^{1/4} \asymp d^{1/2}$, together with a factor $(1 + X/(dH^{1/2}))^\theta$ from possible exceptional eigenvalues. The proof has six steps.

- (1) Poisson summation in ℓ writes the Type I sum as $\sum_{h \neq 0} W_h$. Here each W_h is a smoothed Weyl sum of the exponentials $e(h\nu/\mu)$, where ν runs over the roots of $E\nu^2 + H \equiv 0$ modulo μ . Only the frequencies $|h| \leq (qX)^\eta K/X$ matter, where K is the size of the moduli.
- (2) Following [6], $W_h = \sum_i P_h(z_i)$, where P_h is a Poincaré series of frequency h on $\Gamma_0(q)$ and the z_i are Heegner points (Lemma 9.2).
- (3) For each dyadic range $h \asymp M$, the spectral expansion gives $\sum_h \psi_0(h/M) W_h = \sum_j B_j U_j +$ (Eisenstein), where $U_j = \sum_i u_j(z_i)$ and $B_j = \sum_h \psi_0(h/M) \overline{\rho_j(h)} \check{g}_h(t_j)$, with $\check{g}_h$ a K -Bessel transform of the weight.
- (4) By Cauchy–Schwarz, these sums are bounded in terms of a *Heegner side* $\sum_j |U_j|^2$ and a *frequency side* $\sum_j |B_j|^2$, over dyadic ranges of the spectral parameter.
- (5) The frequency side is bounded with the large sieve inequality of Deshouillers and Iwaniec. Since the argument of the Bessel functions is $\ll (qX)^\eta$, only spectral parameters $|t_j| \leq (qX)^{O(\eta)}$ matter, and the bound obtained is the diagonal term of the large sieve. For an exceptional eigenvalue $\frac{1}{4} - \theta_j^2$ the transform $\check{g}_h(t_j)$ is larger by a factor V^{θ_j} , where $V \approx X/\sqrt{N}$ at the decisive frequencies. Pascadi’s inequality, extended to smooth weights (Lemma 10.5), applies to the sequences that arise and absorbs $V^{2\theta_j}$ when $V \leq \max(M, q)$. The inequality [4, Theorem 5] for general sequences requires $V \leq \max(1, q/M)$, and leads to Theorem 1.3.
- (6) The Heegner side is bounded by the pre-trace formula in terms of the number $\mathcal{P}$ of pairs of Heegner points at bounded hyperbolic distance on $\Gamma_0(q) \backslash \mathbb{H}$. An elementary count gives $\mathcal{P} \ll N^{1+\eta}$ (Lemma 8.4).

Kuznetsov’s formula [20] enters through the inequalities of [4] and [23], which are derived from it.

In the resulting asymptotic formula for the weighted count, the main term is $\gg x^{1/2}d^{-1/2-\varepsilon}$ and the error term is $x^{1/4+\theta/2+\varepsilon}d^{3/4-3\theta/2}$ (Proposition 7.1). This allows $d \leq x^\varpi$ for every $\varpi < (1-2\theta)/(5-6\theta)$, and Theorem 1.1 follows.

Section 13 discusses other approaches and the limits of the method. Treated with Weil's bound for Kloosterman sums, as in Hooley's work on quadratic polynomials [14], the Type I sums were handled in an earlier version of this work for $d \leq x^{1/12-\varepsilon}$, which gives $\delta < 1/24$. The bounds of Duke, Friedlander and Iwaniec [7] for Weyl sums for quadratic roots, uniform in the discriminant, would not improve on this in our range (Remark 13.4). In another earlier version, a bound of the same strength as Theorem 6.1(a) was deduced from a bound of Grimmelt and Merikoski for weighted averages of automorphic kernels [12, Theorem 8.1]; the two arguments correspond closely (Remark 13.5), but the present proof does not use [12]. In both spectral arguments the bottleneck is the bound for $\mathcal{P}$, which ignores the level structure; Remark 13.1 explains what an improvement would give.

1.2. Notation. We write $e(t) = e^{2\pi it}$, $\hat{\psi}(\xi) = \int_{\mathbb{R}} \psi(t)e(-\xi t) dt$, and $\|t\|$ for the distance from $t \in \mathbb{R}$ to the nearest integer. We write $n \sim N$ for $N < n \leq 2N$. The letter χ denotes the non-principal character modulo 4, and

$$r(n) = \#\{(x, y) \in \mathbb{Z}^2 : x^2 + y^2 = n\} = 4 \sum_{k|n} \chi(k) \quad (n \geq 1).$$

For a prime ℓ , v_ℓ is the ℓ -adic valuation. The functions $\tau(n)$ and $\omega(n)$ count the divisors and the distinct prime factors of n , and $\mathbf{1}\{\cdot\}$ is an indicator function. Throughout, a and b are fixed, θ is a fixed admissible exponent, and $\varepsilon, \eta > 0$ are small. Implied constants may depend on a, b, ε, η and on the fixed smooth functions F, W and ψ_0 introduced below, and in Sections 6 and 8–12 on the constants C_ν in the hypothesis (H) of §6, but on nothing else unless indicated.

1.3. Earlier versions. This paper is the final version of a project whose earlier stages are kept in an accompanying repository, together with the scripts for their numerical checks. In order, they are: a treatment of three consecutive integers by Hooley's method and Weil's bound, which gives $\delta < 1/52$; its extension to all patterns, which gives $\delta < 1/24$ (Remark 13.4); a version in which Theorem 6.1(a) is deduced from [12, Theorem 8.1] (Remark 13.5); a line-by-line verification of that theorem; and a version based on the Kuznetsov formula, which compares the two spectral approaches. The present text is self-contained. It does not rely on those documents, and the proof of Theorem 1.1 given here does not use [12, 13].

2. THE FAMILIES

Lemma 2.1. *If a and b are both even, then $S_{a,b}(x) \geq S_{a/2,b/2}(x/2)$.*

Proof. If $1 \leq n' \leq x/2$ and $n', n' + a/2, n' + b/2 \in \mathcal{S}$, then $n = 2n' \leq x$ and $n, n + a, n + b \in \mathcal{S}$, because $2\mathcal{S} \subseteq \mathcal{S}$. $\square$

Definition 2.2. A pair $(B, A) \in \mathbb{Z}^2$ is *admissible* if $B > 0$ is odd, $A \notin \{0, B\}$, and neither

$$B \equiv 1, A \equiv 3 \pmod{4} \quad \text{nor} \quad B \equiv 3, A \equiv 2 \pmod{4}.$$

Lemma 2.3. *Let $0 < a < b$ with a or b odd. There are $p_0 \in \{0, a\}$ and an admissible pair (B, A) such that*

$$\{n + p_0, n + p_0 + B, n + p_0 + A\} = \{n, n + a, n + b\} \quad \text{for all } n \in \mathbb{Z}.$$

One such choice is given in Table 2.

case	p_0	B	A	automatic pair	detected
1 a odd, b even, $(a, b) \not\equiv (3, 2) \pmod{4}$	0	a	b	$n, n + a$	$n + b$
2 $(a, b) \equiv (3, 2) \pmod{4}$	a	$b - a$	$-a$	$n + a, n + b$	n
3 a even, b odd, $(a, b) \not\equiv (2, 3) \pmod{4}$	0	b	a	$n, n + b$	$n + a$
4 $(a, b) \equiv (2, 3) \pmod{4}$	a	$b - a$	$-a$	$n + a, n + b$	n
5 a, b odd, $(a, b) \not\equiv (1, 3) \pmod{4}$	0	a	b	$n, n + a$	$n + b$
6 $(a, b) \equiv (1, 3) \pmod{4}$	0	b	a	$n, n + b$	$n + a$

TABLE 2. The choice of (p_0, B, A) in Lemma 2.3.

Proof. The six rows cover all pairs (a, b) with a or b odd. In each row $B > 0$ is odd, and the three numbers $n + p_0, n + p_0 + B, n + p_0 + A$ are $n, n + a, n + b$ in some order, so $A \notin \{0, B\}$. In rows 1, 3 and 5 the excluded congruences are exactly those that would violate admissibility. In rows 2, 4 and 6 we have $(B, A) \equiv (3, 1), (1, 2)$ and $(3, 1) \pmod{4}$ respectively, and these pairs are admissible. $\square$

From now on we assume, as we may by Lemma 2.1, that a or b is odd, and we fix p_0, B and A as in Lemma 2.3. For $n \in \mathbb{Z}$ we write

$$(2.1) \quad P = n + p_0, \quad T = P + A, \quad Q = 2P + B,$$

so that $\{P, P + B, T\} = \{n, n + a, n + b\}$. The numbers P and $P + B$ form the automatic pair, and T is the detected element. Put

$$(2.2) \quad c = 2A - B, \quad m_d = d^2 + 2cd + B^2 = (d + c)^2 + 4A(B - A).$$

For example, for $\{0, 1, 2\}$ Table 2 gives $p_0 = 0, (B, A) = (1, 2)$ and $m_d = d^2 + 6d + 1$.

Lemma 2.4. *Let $d, u, T \in \mathbb{Z}$ satisfy $4dT = u^2 + m_d$, and put $P = T - A$ and $Q = 2P + B$. Then $2dQ = u^2 + d^2 + B^2$ and*

$$4P(P + B) = Q^2 - B^2 = (Q - d)^2 + u^2.$$

Proof. We have $Q = 2T - c$, so $2dQ = 4dT - 2cd = u^2 + m_d - 2cd = u^2 + d^2 + B^2$. Next, $Q - B = 2P$ and $Q + B = 2(P + B)$. Finally, $(Q - d)^2 + u^2 = Q^2 - 2dQ + d^2 + u^2 = Q^2 - B^2$. $\square$

Lemma 2.5. *In Lemma 2.4, suppose that d is a prime with $d \equiv 1 \pmod{4}$ and that $P \geq 1$. Then $P \in \mathcal{S}$ and $P + B \in \mathcal{S}$.*

Proof. Recall that $z \geq 1$ lies in $\mathcal{S}$ if and only if $v_\ell(z)$ is even for every prime $\ell \equiv 3 \pmod{4}$. By Lemma 2.4, $4P(P + B) \in \mathcal{S}$, so $P(P + B) \in \mathcal{S}$. Let $\ell \equiv 3 \pmod{4}$ be prime. If ℓ divides at most one of P and $P + B$, then $v_\ell(P)$ and $v_\ell(P + B)$ are 0 or $v_\ell(P(P + B))$, hence even. If ℓ divides both, then $\ell \mid B$ and $\ell \mid Q = 2P + B$, so $\ell \mid u^2 + d^2$ by Lemma 2.4. Since -1 is not a square modulo ℓ , this forces $\ell \mid d$, which is impossible. $\square$

Lemma 2.6. *Let n be an integer with $P = n + p_0 \geq 1$, and let T be as in (2.1). Then*

$$\#\{(d, u) \in \mathbb{Z}^2 : d \geq 1, 4dT = u^2 + m_d\} \leq r(P(P + B)).$$

Proof. The number $Q = 2P + B$ depends only on n . By Lemma 2.4, the map $(d, u) \mapsto (Q - d, u)$ sends the set on the left injectively into the set of representations of $4P(P + B)$ as a sum of two squares. Finally, $r(4z) = r(z)$. $\square$

3. LOCAL ANALYSIS AT THE PRIME 2

Let $\mathcal{N} \subset \mathbb{Z}_2$ be the set of non-zero 2-adic integers whose odd part is $\equiv 1 \pmod{4}$. Then $\mathcal{S} \setminus \{0\} \subseteq \mathcal{N}$, since an odd element of $\mathcal{S}$ is $\equiv 1 \pmod{4}$. We write $z = 2^{v_2(z)}z'$ with z' odd. Recall that a non-zero $z \in \mathbb{Q}_2$ is a square if and only if $v_2(z)$ is even and $z' \equiv 1 \pmod{8}$.

Lemma 3.1. *Let (B, A) be admissible. There is $n_* \in \mathbb{Z}$ such that*

- (i) $Q_* := 2n_* + B \equiv 1 \pmod{4}$;
- (ii) $n_*(n_* + B)$ is a non-zero square in $\mathbb{Q}_2$;
- (iii) $T_* := n_* + A \in \mathcal{N}$.

Proof. Let $\sigma = 1$ if $B \equiv 1 \pmod{4}$ and $\sigma = -1$ if $B \equiv 3 \pmod{4}$, so that $\sigma B \equiv 1 \pmod{4}$. For an integer $s \geq 1$ and an odd integer o put

$$n = 4^s o \quad (\sigma = 1), \quad n = 4^s o - B \quad (\sigma = -1).$$

Then $Q = 2n + B$ equals $2 \cdot 4^s o + B$, respectively $2 \cdot 4^s o - B$, so $Q \equiv \sigma B \equiv 1 \pmod{4}$ and (i) holds. Moreover $n(n + B) = 4^s o(4^s o + \sigma B)$, and since $o^2 \equiv 1 \pmod{8}$,

$$o(4^s o + \sigma B) \equiv 4^s + \sigma B o \pmod{8}.$$

Hence (ii) holds if and only if $o \equiv c_s \pmod{8}$, where $c_s \equiv \sigma B + 4 \pmod{8}$ if $s = 1$ and $c_s \equiv \sigma B \pmod{8}$ if $s \geq 2$. In both cases $c_s \equiv 1 \pmod{4}$. Finally $T = n + A = 4^s o + C$, where $C = A$ if $\sigma = 1$ and $C = A - B$ if $\sigma = -1$. We have $C \neq 0$, because $A \notin \{0, B\}$. It remains to choose s , and the class of o modulo a power of 2 compatible with $o \equiv c_s \pmod{8}$, so that $4^s o + C \in \mathcal{N}$.

C odd. We claim that $C \equiv 1 \pmod{4}$. If $\sigma = 1$, then $C = A$ is odd, and admissibility excludes $A \equiv 3 \pmod{4}$. If $\sigma = -1$, then $C = A - B$ is odd, so A is even; admissibility excludes $A \equiv 2 \pmod{4}$, so $A \equiv 0 \pmod{4}$ and $C \equiv -B \equiv 1 \pmod{4}$. We take $s = 2$ and $o = c_2$; then $T \equiv C \equiv 1 \pmod{4}$.

C even. Write $C = 2^t w$ with $t \geq 1$ and w odd.

- If $w \equiv 1 \pmod{4}$, take $s = \max(2, \lceil (t+2)/2 \rceil)$ and $o = c_s$. Then $T = 2^t(w + 2^{2s-t}o)$ with $2s - t \geq 2$, so $w + 2^{2s-t}o \equiv 1 \pmod{4}$.
- If $w \equiv 3 \pmod{4}$ and t is odd, take $s = (t+1)/2$ and $o = c_s$. Then $T = 2^t(w + 2o)$ and $w + 2o \equiv 3 + 2 \equiv 1 \pmod{4}$.
- If $w \equiv 3 \pmod{4}$ and t is even, take $s = t/2 \geq 1$. Then $T = 2^t(w + o)$, and $w + c_s \equiv 0 \pmod{4}$. If $w + c_s \equiv 4 \pmod{8}$, take $o \equiv 4 - w \pmod{16}$; then $o \equiv c_s \pmod{8}$ and $w + o \equiv 4 \pmod{16}$. If $w + c_s \equiv 0 \pmod{8}$, take $o \equiv 8 - w \pmod{32}$; then $o \equiv c_s \pmod{8}$ and $w + o \equiv 8 \pmod{32}$. In both cases $w + o \in \mathcal{N}$.

In every case $T \in \mathcal{N}$. Taking for o a positive integer in the required class gives $n_* \in \mathbb{Z}$ with (i)–(iii). $\square$

Proposition 3.2. *Let (B, A) be admissible and let m_d be as in (2.2). There are integers $v \geq 2$ and $d_0 \equiv 1 \pmod{4}$ with the following property. Put $J = v + 2$ and $j = \lceil (v+2)/2 \rceil$. For all integers $d \equiv d_0 \pmod{2^J}$ and $u \equiv 0 \pmod{2^j}$,*

$$v_2(u^2 + m_d) = v \quad \text{and} \quad 2^{-v}(u^2 + m_d) \equiv d \pmod{4}.$$

Consequently, if in addition $d \mid u^2 + B^2$, then $T = (u^2 + m_d)/(4d)$ is an integer of the form $2^{v-2}T'$ with $T' \equiv 1 \pmod{4}$.

Proof. Let n_* , Q_* and T_* be as in Lemma 3.1. By (ii) there is $t_* \in \mathbb{Z}_2$ with $t_*^2 = 4n_*(n_* + B)$, and $v_2(n_*(n_* + B))$ is even. It is also positive, because B is odd and so exactly one of n_* and $n_* + B$ is even. Hence $v_2(t_*) \geq 2$. Put $d_* = Q_* - t_*$. Then $d_* \equiv Q_* \equiv 1 \pmod{4}$ and

$$d_*(Q_* + t_*) = Q_*^2 - t_*^2 = (2n_* + B)^2 - 4n_*(n_* + B) = B^2.$$

Hence $d_*^2 - 2d_*Q_* + B^2 = d_*(d_* - 2Q_* + Q_* + t_*) = 0$, that is, $2d_*Q_* = d_*^2 + B^2$. This is the relation of Lemma 2.4 with $u = 0$, read 2-adically, and it gives

$$m_{d_*} = d_*^2 + 2cd_* + B^2 = 2d_*(Q_* + c) = 4d_*(n_* + A) = 4d_*T_*.$$

Let $v = v_2(m_{d_*}) = 2 + v_2(T_*) \geq 2$. By (iii), $2^{-v}m_{d_*} = d_*T'_* \equiv d_* \pmod{4}$. Choose $d_0 \in \mathbb{Z}$ with $d_0 \equiv d_* \pmod{2^J}$; then $d_0 \equiv 1 \pmod{4}$. If $d \equiv d_0 \pmod{2^J}$, then $m_d \equiv m_{d_*} \pmod{2^J}$, because m_d is a polynomial in d with integer coefficients. If moreover $u \equiv 0 \pmod{2^j}$, then $u^2 \equiv 0 \pmod{2^J}$ since $2j \geq v + 2 = J$. Hence $u^2 + m_d \equiv m_{d_*} \pmod{2^J}$, and since $v_2(m_{d_*}) = v = J - 2$ we get $v_2(u^2 + m_d) = v$ and $2^{-v}(u^2 + m_d) \equiv 2^{-v}m_{d_*} \equiv d_* \equiv d \pmod{4}$.

For the last assertion note that $m_d \equiv B^2 \pmod{d}$, so $d \mid u^2 + m_d$. Since $d \equiv 1 \pmod{4}$ is odd and $4 \mid u^2 + m_d$, T is an integer with $v_2(T) = v - 2$, and its odd part T' satisfies $dT' = 2^{-v}(u^2 + m_d) \equiv d \pmod{4}$, that is, $T' \equiv 1 \pmod{4}$. $\square$

4. GOOD PRIMES AND THE DEDUCTION OF THE MAIN THEOREMS

We fix v and d_0 as in Proposition 3.2 and put

$$(4.1) \quad J = v + 2, \quad j = \lceil (v + 2)/2 \rceil, \quad E = 2^{2j-v} \in \{4, 8\}.$$

Note that $j \leq v$, because $v \geq 2$. Let $d_1 = 8(|c| + B)$. For $d > d_1$ we have $d > B$ and $|2cd + B^2| \leq d^2/2$, so $\frac{1}{2}d^2 \leq m_d \leq 2d^2$.

Definition 4.1. A prime d is *good* if $d \equiv d_0 \pmod{2^J}$, $d > d_1$, and neither m_d nor $m_d/3$ is a perfect square.

Let d be a good prime. Then $d \equiv 1 \pmod{4}$ and $d \nmid B$. Since $m_d \equiv B^2 \pmod{d}$, also $d \nmid m_d$. By Proposition 3.2 with $u = 0$, $v_2(m_d) = v$, so

$$H_d := 2^{-v}m_d$$

is an odd positive integer with $d \nmid H_d$. Since $2^v E = 2^{2j}$, we have

$$(4.2) \quad (2^j \ell)^2 + m_d = 2^v (E \ell^2 + H_d) \quad (\ell \in \mathbb{Z}).$$

Moreover $N_d := EH_d = 4^{j-v}m_d$, and since $j \leq v$, N_d is neither a square nor three times a square: if $N_d = s^2$ or $N_d = 3s^2$, then $m_d = (2^{v-j}s)^2$ or $m_d = 3(2^{v-j}s)^2$.

Fix a smooth $F : \mathbb{R} \rightarrow [0, 1]$ supported in $[1/2, 1]$ and not identically zero, and put $c_F = \int_0^1 F(t)t^{-1/2} dt > 0$. For a good prime d and $x \geq 1$ define

$$(4.3) \quad \Sigma_d(x) = \sum_{\substack{\ell \in \mathbb{Z} \\ d \mid E\ell^2 + H_d}} r(T_d(\ell)) F\left(\frac{T_d(\ell)}{x}\right), \quad T_d(\ell) = \frac{2^{v-2}(E\ell^2 + H_d)}{d} = \frac{(2^j \ell)^2 + m_d}{4d}.$$

By (4.2), and since $m_d \equiv B^2 \pmod{d}$, we have $d \mid E\ell^2 + H_d$ if and only if $d \mid u^2 + B^2$, where $u = 2^j \ell$. Proposition 3.2 therefore shows that for every ℓ in (4.3) the number $T_d(\ell)$ is an integer of the form $2^{v-2}T'$ with $T' \equiv 1 \pmod{4}$.

Proposition 4.2. Let $\mathcal{D}$ be a finite set of good primes and $x \geq x_0$. Then

$$\sum_{d \in \mathcal{D}} \Sigma_d(x) \leq S_{a,b}(x) \cdot \max_{n \leq x} r(T) r(P(P+B)) \ll_\varepsilon S_{a,b}(x) x^\varepsilon,$$

with P and T given by (2.1).

Proof. Let $d \in \mathcal{D}$, and let ℓ be such that the corresponding term in (4.3) is non-zero. Put $u = 2^j \ell$ and $T = T_d(\ell)$. Then T is an integer, $4dT = u^2 + m_d$, $x/2 \leq T \leq x$ and $T \in \mathcal{S}$. Put $P = T - A$ and $n = P - p_0$. Then $\{n, n+a, n+b\} = \{P, P+B, T\}$, and since n is the smallest of these numbers, $T - b \leq n \leq T$. Hence $1 \leq n \leq x$, and $P \geq n \geq 1$, for $x \geq x_0$. By Lemmas 2.4 and 2.5,

P and $P+B$ lie in $\mathcal{S}$. So n is counted by $S_{a,b}(x)$. By Lemma 2.6, each such n arises from at most $r(P(P+B))$ pairs (d, u) , and u determines ℓ . The last bound follows from $r(z) \leq 4\tau(z) \ll z^\varepsilon$. $\square$

Theorem 4.3. *Let $0 < \varpi < \varpi_\theta := (1 - 2\theta)/(5 - 6\theta)$. There is $\varepsilon_0 = \varepsilon_0(\varpi, \theta) > 0$ such that for $0 < \varepsilon \leq \varepsilon_0$ and $x \geq x_0(a, b, \varpi, \varepsilon)$,*

$$\Sigma_d(x) \gg_\varepsilon x^{1/2} d^{-1/2-\varepsilon}$$

for every good prime $d \leq x^\varpi$. The implied constant is ineffective.

The proof in §7 uses Theorem 6.1(a). With Theorem 6.1(b) in its place, the same proof gives the conclusion of Theorem 4.3 for every $0 < \varpi < \varpi'_\theta := (1 - 2\theta)/(5 - 4\theta)$, without using [23, Theorem 2].

Deduction of Theorems 1.1 and 1.3. Let $\delta < \delta_\theta = \varpi_\theta/2$. Choose ϖ with $2\delta < \varpi < \varpi_\theta$, and then $\varepsilon \leq \varepsilon_0(\varpi, \theta)$ with $\varpi(\frac{1}{2} - \varepsilon) - 2\varepsilon > \delta$. By Lemma 2.1, applied repeatedly with x replaced by $x/2^i$, we may assume that a or b is odd.

The primes $d \leq x^\varpi$ with $d \equiv d_0 \pmod{2^J}$ that are not good are of three kinds. There are $O(1)$ primes $d \leq d_1$. There are $O(1)$ primes with $m_d = s^2$, because $(d+c)^2 - s^2 = -4A(B-A) \neq 0$ has finitely many solutions. Finally, there are the primes with $m_d = 3s^2$. The equation $(d+c)^2 - 3s^2 = -4A(B-A)$ has $O(\log x)$ solutions with $0 < d \leq x$, because its solutions form finitely many orbits under the units of $\mathbb{Z}[\sqrt{3}]$.

Let $\mathcal{D}$ be the set of good primes $d \leq x^\varpi$. The prime number theorem in arithmetic progressions gives $\sum_{d \in \mathcal{D}} d^{-1/2-\varepsilon} \gg x^{\varpi(1/2-\varepsilon)} (\log x)^{-1}$. By Theorem 4.3 and Proposition 4.2,

$$S_{a,b}(x) \gg x^{-\varepsilon} \sum_{d \in \mathcal{D}} \Sigma_d(x) \gg x^{1/2+\varpi(1/2-\varepsilon)-\varepsilon} (\log x)^{-1} \gg x^{1/2+\delta}.$$

This proves Theorem 1.1. The same argument with ϖ'_θ and $\delta'_\theta = \varpi'_\theta/2$ in place of ϖ_θ and δ_θ proves Theorem 1.3. $\square$

For $\theta = 7/64$ we get $\varpi_\theta = 25/139$ and $\delta_\theta = 25/278$, and $\varpi'_\theta = 25/146$ and $\delta'_\theta = 25/292$. For $\theta = 1/4$ we get $\varpi_\theta = 1/7$, $\delta_\theta = 1/14$, $\varpi'_\theta = 1/8$ and $\delta'_\theta = 1/16$, and for $\theta = 0$ we get $\varpi_\theta = \varpi'_\theta = 1/5$ and $\delta_\theta = \delta'_\theta = 1/10$. This proves Corollary 1.2 and gives Table 1. The rest of the paper proves Theorem 4.3.

5. ARITHMETIC PRELIMINARIES

5.1. A smoothed hyperbola identity. Fix a smooth non-increasing $\lambda_0 : \mathbb{R} \rightarrow [0, 1]$ with $\lambda_0(t) = 1$ for $t \leq -\log 2$ and $\lambda_0(t) = 0$ for $t \geq \log 2$. Put $\lambda(t) = \frac{1}{2}(\lambda_0(t) + 1 - \lambda_0(-t))$ and $W(t) = \lambda(\log t)$ for $t > 0$. Then W is smooth with values in $[0, 1]$, and

$$(5.1) \quad W(t) = 1 \ (t \leq \tfrac{1}{2}), \quad W(t) = 0 \ (t \geq 2), \quad W(t) + W(1/t) = 1.$$

Lemma 5.1. *Let $s \geq 0$ be an integer and $T = 2^s T'$ with $T' \equiv 1 \pmod{4}$. Then*

$$r(T) = 8 \sum_{k|T} \chi(k) W(2^{s/2} k / \sqrt{T}).$$

Proof. First, $r(T) = r(T') = 4 \sum_{k|T'} \chi(k)$. By (5.1),

$$\sum_{k|T'} \chi(k) = \sum_{k|T'} \chi(k) W(k/\sqrt{T'}) + \sum_{k|T'} \chi(k) W(\sqrt{T'}/k).$$

In the second sum substitute $k = T'/i$ and use $\chi(T'/i) = \chi(i)$, which holds because $T' \equiv 1 \pmod{4}$. So the two sums are equal. Finally, the odd divisors of T are those of T' , χ vanishes on even numbers, and $\sqrt{T'} = 2^{-s/2} \sqrt{T}$. $\square$

5.2. Roots of $\nu^2 \equiv -m_d$. Throughout this subsection and the next, d is a good prime, $m = m_d$ and $H = H_d$. For $n \geq 1$ let

$$\varrho(n) = \#\{\nu \bmod n : \nu^2 + m \equiv 0 \pmod{n}\},$$

a multiplicative function.

Lemma 5.2. (a) For odd n , $\varrho(n) = \#\{t \bmod n : Et^2 + H \equiv 0 \pmod{n}\}$.

(b) Let ℓ be an odd prime and $i \geq 1$. If $\ell \nmid m$ then $\varrho(\ell^i) = 1 + \left(\frac{-m}{\ell}\right)$. In general $\varrho(\ell^i) \leq 2\ell^{\min(i, v_\ell(m))/2}$. Consequently $\varrho(n) \leq 2^{\omega(n)} \gcd(n, m)^{1/2} \ll_\varepsilon n^\varepsilon d$ for odd n .

(c) $\varrho(d^i) = 2$ for all $i \geq 1$.

Proof. (a) This follows from (4.2), since 2^v and 2^j are invertible modulo n .

(b) For $\ell \nmid m$ use Hensel's lemma. Let $\alpha = v_\ell(m) \geq 1$. For $i \leq \alpha$ the congruence reads $\nu^2 \equiv 0 \pmod{\ell^i}$, which has $\ell^{\lfloor i/2 \rfloor}$ solutions. For $i > \alpha$ a solution has $v_\ell(\nu) = \alpha/2$, so there are none unless α is even. In that case $\nu = \ell^{\alpha/2} \nu_1$, where ν_1 is taken modulo $\ell^{i-\alpha/2}$ and satisfies $\nu_1^2 \equiv -m\ell^{-\alpha} \pmod{\ell^{i-\alpha}}$. There are at most 2 such ν_1 modulo $\ell^{i-\alpha}$, each with $\ell^{\alpha/2}$ lifts. The consequence follows by multiplicativity and $m \leq 2d^2$.

(c) Since $m \equiv B^2 \pmod{d}$ with $d \nmid B$ and $d \equiv 1 \pmod{4}$, we get $\varrho(d^i) = 1 + \left(\frac{-B^2}{d}\right) = 2$. $\square$

For odd k put $f(k) = \varrho(dk)/2$, and put $f(k) = 0$ for even k . Then f is multiplicative, with

$$(5.2) \quad f(d^i) = 1, \quad f(\ell^i) = \varrho(\ell^i) \quad (\ell \neq d).$$

5.3. The singular series. Let $\psi(n) = \left(\frac{m}{n}\right)$ (Jacobi symbol) for odd $n \geq 1$, and $\psi(n) = 0$ for even n .

Lemma 5.3. (a) ψ is a non-principal real Dirichlet character modulo $8m$.

(b) For $t \geq 1$, $A_f(t) := \sum_{k \leq t} \chi(k)f(k) \ll_\varepsilon t^{1/2+\varepsilon} d^{1+\varepsilon}$.

(c) The series $\mathfrak{L}_d := \sum_{k \geq 1} \chi(k)f(k)/k$ converges, $|\mathfrak{L}_d| \ll d^{1+\varepsilon}$, and

$$\mathfrak{L}_d = L(1, \chi) L(1, \psi) \mathcal{P}_d \quad \text{with} \quad \mathcal{P}_d \geq \frac{6}{\pi^2} \cdot \frac{\phi(dm)}{dm}.$$

(d) $\mathfrak{L}_d \gg_\varepsilon d^{-\varepsilon}$.

Proof. (a) Write $m = 2^t m'$ with m' odd. For odd $n > 0$, quadratic reciprocity for Jacobi symbols gives

$$\left(\frac{m}{n}\right) = \left(\frac{2}{n}\right)^t (-1)^{\frac{m'-1}{2} \frac{n-1}{2}} \left(\frac{n}{m'}\right),$$

which is a character modulo $8m' \mid 8m$. It is non-principal because m is not a square. If m' is not a square, then $\left(\frac{\cdot}{m'}\right)$ is non-principal, and the Chinese remainder theorem gives $n \equiv 1 \pmod{8}$ with $\psi(n) = \left(\frac{n}{m'}\right) = -1$. If $m' = s^2$, then t is odd. Choose $n \equiv 3 \pmod{8}$ with $n \equiv 1 \pmod{s}$; then $\psi(n) = (-1)^t \cdot (-1)^{\frac{m'-1}{2} \frac{n-1}{2}} \left(\frac{n}{s}\right)^2 = -1$, because t is odd and $m' = s^2 \equiv 1 \pmod{8}$.

(b) We compare the Euler factors of χf and $\chi * \psi$, writing $Y = \ell^{-s}$.

- At $\ell = 2$, both χf and $\chi * \psi$ have factor 1.
- For odd $\ell \nmid dm$, (5.2) and Lemma 5.2(b) give, with $\psi(\ell) = \chi(\ell) \left(\frac{-m}{\ell}\right)$,

$$\sum_{i \geq 0} \chi(\ell^i) f(\ell^i) Y^i = 1 + \left(1 + \left(\frac{-m}{\ell}\right)\right) \frac{\chi(\ell) Y}{1 - \chi(\ell) Y} = \frac{1 - Y^2}{(1 - \chi(\ell) Y)(1 - \psi(\ell) Y)}.$$

- At $\ell = d$, $\chi(d) = 1$ because $d \equiv 1 \pmod{4}$, and $\psi(d) = \left(\frac{B^2}{d}\right) = 1$. The factor of χf is $(1 - Y)^{-1} = (1 - Y) \cdot (1 - Y)^{-2}$.

- For odd $\ell \mid m$, $\psi(\ell) = 0$. The factor of χf is $P_\ell(Y) := \sum_i \chi(\ell)^i \varrho(\ell^i) Y^i = (1 - \chi(\ell)Y)P_\ell(Y) \cdot (1 - \chi(\ell)Y)^{-1}$.

Hence $\chi f = (\chi * \psi) * \beta$, where β is multiplicative, supported on odd integers, and given as follows: $\beta(\ell^2) = -1$ and $\beta(\ell^i) = 0$ for $i \neq 0, 2$ when $\ell \nmid 2dm$; $\beta(d) = -1$ and $\beta(d^i) = 0$ for $i \geq 2$; and $\beta(\ell^i) = \chi(\ell)^i (\varrho(\ell^i) - \varrho(\ell^{i-1}))$ for odd $\ell \mid m$. By Lemma 5.2(b), $|\beta(\ell^i)| \leq 2\ell^{\min(i, \alpha)/2}$ for odd $\ell \mid m$ with $\alpha = v_\ell(m)$, so

$$\sum_{i \geq 0} |\beta(\ell^i)| \ell^{-i(1/2+\varepsilon)} \leq 1 + 2\alpha + 2 \sum_{i \geq 1} \ell^{-i/2} \leq 2\alpha + 5.$$

Therefore $\sum_n |\beta(n)| n^{-1/2-\varepsilon} \leq \zeta(1+2\varepsilon)(1+d^{-1/2}) \prod_{\ell \mid m, \ell \text{ odd}} (2v_\ell(m) + 5) \ll_\varepsilon d^\varepsilon$.

The Pólya–Vinogradov inequality [3, Ch. 23], applied to the primitive character inducing ψ together with Möbius inversion over the remaining primes dividing $8m$, gives $\Psi^* \ll_\varepsilon m^{1/2+\varepsilon}$, where $\Psi^* := \sup_y |\sum_{n \leq y} \psi(n)|$. The hyperbola method with parameter $\sqrt{y}$ gives $|\sum_{n \leq y} (\chi * \psi)(n)| \leq 3\sqrt{y}(\Psi^* + 1)$ for $y \geq 1$. Finally $A_f(t) = \sum_{i \leq t} \beta(i) \sum_{n \leq t/i} (\chi * \psi)(n)$, so

$$|A_f(t)| \ll m^{1/2+\varepsilon} \sum_{i \leq t} |\beta(i)| (t/i)^{1/2} \leq m^{1/2+\varepsilon} t^{1/2+\varepsilon} \sum_i |\beta(i)| i^{-1/2-\varepsilon} \ll t^{1/2+\varepsilon} d^{1+3\varepsilon}.$$

Renaming ε gives (b).

(c) Partial summation and (b) give convergence and $|\mathfrak{L}_d| \leq \int_1^\infty |A_f(t)| t^{-2} dt \ll d^{1+\varepsilon}$. For $\text{Re } s > 1$ we have $\sum_k \chi(k) f(k) k^{-s} = L(s, \chi) L(s, \psi) \mathcal{P}_d(s)$ with $\mathcal{P}_d(s) = \sum_n \beta(n) n^{-s}$. Both sides are analytic for $\text{Re } s > 1/2$: the left-hand side because it converges there by (b), and the right-hand side because χ and ψ are non-principal and $\mathcal{P}_d(s)$ converges absolutely. So the identity extends to $s = 1$, with $\mathcal{P}_d := \mathcal{P}_d(1) = \prod_\ell \beta_\ell(\ell^{-1})$, where $\beta_\ell(Y) = \sum_i \beta(\ell^i) Y^i$.

The Euler factors are $1 - \ell^{-2}$ for $\ell \nmid 2dm$, $1 - d^{-1}$ at $\ell = d$, and $(1 - \chi(\ell)\ell^{-1})P_\ell(\ell^{-1})$ for odd $\ell \mid m$. In the last case, if $\chi(\ell) = 1$ then $P_\ell(\ell^{-1}) \geq 1$. If $\chi(\ell) = -1$, let u be distributed according to Haar measure on $\mathbb{Z}_\ell$. Since $\varrho(\ell^i)\ell^{-i} = \text{Prob}(\ell^i \mid u^2 + m)$, we get

$$P_\ell(\ell^{-1}) = \sum_{i \geq 0} (-1)^i \text{Prob}(\ell^i \mid u^2 + m) = \text{Prob}(v_\ell(u^2 + m) \text{ even}) \geq \text{Prob}(\ell \nmid u^2 + m) = 1 - \frac{1}{\ell},$$

since $\varrho(\ell) = 1$. In all cases the factor is $\geq 1 - \ell^{-1}$, which gives the lower bound for $\mathcal{P}_d$.

(d) We have $L(1, \chi) = \pi/4$. Let ψ^* be the primitive character inducing ψ . Its conductor q^* divides $8m$, and $L(1, \psi) \geq L(1, \psi^*) \phi(8m)/(8m)$. By Siegel's theorem [3, Ch. 21], $L(1, \psi^*) \gg_\varepsilon (q^*)^{-\varepsilon}$. Combine these bounds with (c) and with $\phi(n)/n \gg (\log \log 3n)^{-1}$. $\square$

6. THE TYPE I ESTIMATE

Consider the following hypothesis.

(H) $E \in \{4, 8\}$; $H \geq 1$ is odd; $d \geq 3$ is a prime with $d \nmid H$; $N = EH$ is neither a square nor three times a square; $q = 4Ed$; $\kappa \in \{1, 3\}$; $\Delta \geq 1$, $X \geq N^{1/2}$ and $1 \leq K \leq qX$; $\psi_1, \psi_2 : \mathbb{R} \rightarrow \mathbb{C}$ are smooth, ψ_1 is supported in $[1, 2]$ and ψ_2 in $[-1, 1]$, and $\|\psi_i^{(\nu)}\|_\infty \leq C_\nu \Delta^\nu$ for all $\nu \geq 0$.

Under (H) put $\varrho_{E,H}(\mu) = \#\{\nu \bmod \mu : E\nu^2 + H \equiv 0 \pmod{\mu}\}$ and

$$(6.1) \quad \Xi = \Xi(\psi_1, \psi_2) = \sum_{\mu \equiv \kappa d \pmod{4d}} \psi_1\left(\frac{\mu}{K}\right) \left(\sum_{\substack{\ell \in \mathbb{Z} \\ \mu \mid E\ell^2 + H}} \psi_2\left(\frac{\ell}{X}\right) - \frac{\varrho_{E,H}(\mu)}{\mu} X \int_{\mathbb{R}} \psi_2(t) dt \right).$$

Theorem 6.1. *Assume (H). For every $\eta > 0$ there is $C(\eta)$ such that the following holds.*

(a) *We have*

$$\Xi \ll \Delta^{C(\eta)} (qXH)^\eta X^{1/2} H^{1/4} \left(1 + \frac{X}{dH^{1/2}}\right)^\theta.$$

(b) *We also have*

$$\Xi \ll \Delta^{C(\eta)}(qXH)^\eta X^{1/2} H^{1/4} \left(1 + \frac{K}{dH^{1/2}}\right)^\theta.$$

The implied constants depend only on η and on the constants C_ν in (H) (and on two auxiliary functions that are fixed once and for all in the proof).

Theorem 6.1 is proved in Sections 8–12. Part (b) is proved in the same way as part (a), with Lemma 10.3 in place of Lemma 10.5 in §12.2; its proof uses the inequalities of [4] but not [23, Theorem 2].

For $X < K \leq qX$ the moduli exceed the length of the ℓ -sum. Throughout the range $K \leq qX$, Theorem 6.1 gives square-root cancellation in X , with a loss $H^{1/4}$ that does not depend on K . If $K < (qX)^{-\eta} X$, the sum Ξ is negligible (see the remark after Lemma 9.1).

7. THE ASYMPTOTIC FORMULA FOR $\Sigma_d(x)$

Throughout this section d is a good prime, $m = m_d$, $H = H_d$, $N = EH$, $q = 4Ed$, $0 < \varepsilon < 1/100$, and

$$(7.1) \quad d \leq x^{1/4}, \quad x \geq x_0(a, b, \varepsilon).$$

Proposition 7.1. *Under (7.1),*

$$\Sigma_d(x) = \frac{32 c_F \mathfrak{L}_d}{2^j} \sqrt{\frac{x}{d}} + O(x^{1/4+\varepsilon} d^{1/2+\varepsilon} + x^{1/4+\theta/2+\varepsilon} d^{3/4-3\theta/2}).$$

If Theorem 6.1(b) is used in place of Theorem 6.1(a), the proof gives the same formula with $x^{1/4+\theta/2+\varepsilon} d^{3/4-\theta}$ in place of $x^{1/4+\theta/2+\varepsilon} d^{3/4-3\theta/2}$.

7.1. Set-up. For real t put $T(t) = 2^{v-2}(Et^2 + H)/d$, so that $T(\ell) = T_d(\ell)$, and let $X > 0$ be defined by $T(X) = x$. Since $T(0) = m/(4d) \leq d/2$,

$$(7.2) \quad \frac{dx}{2^{v-1}E} \leq X^2 \leq \frac{dx}{2^{v-2}E}.$$

In particular $X \asymp (dx)^{1/2}$, and $X \geq N^{1/2}$ because $N \leq 2Ed^2$. For $y > 0$ and $t \in \mathbb{R}$ put

$$\Phi_y(t) = F(T(t)/x) W\left(\frac{2^{(v-2)/2}y}{\sqrt{T(t)}}\right).$$

Let $\ell \in \mathbb{Z}$ with $d \mid E\ell^2 + H$. By §4, $T(\ell) = 2^{v-2}T'$ with $T' \equiv 1 \pmod{4}$, so Lemma 5.1 applies with $s = v-2$. For odd k we have $k \mid T(\ell)$ if and only if $k \mid T'$, that is, if and only if $dk \mid E\ell^2 + H$, since $E\ell^2 + H = dT'$. Hence

$$(7.3) \quad \Sigma_d(x) = 8 \sum_{k \text{ odd}} \chi(k) \sum_{\substack{\ell \in \mathbb{Z} \\ dk \mid E\ell^2 + H}} \Phi_k(\ell).$$

Only $k < k_+ := 2^{1-(v-2)/2} \sqrt{x}$ contribute, because $W(t) = 0$ for $t \geq 2$ and $T(t) \leq x$ on the support of $F(T(t)/x)$. We put

$$(7.4) \quad \mathcal{M}_d = 8 \sum_{k \text{ odd}} \chi(k) \frac{\varrho(dk)}{dk} \int_{\mathbb{R}} \Phi_k(t) dt, \quad \mathcal{E}_d = \Sigma_d(x) - \mathcal{M}_d.$$

By Lemma 5.2(a), $\varrho(dk) = \varrho_{E,H}(dk)$ for odd k .

7.2. The main term.

Lemma 7.2. $\mathcal{M}_d = \frac{32 c_F \mathfrak{L}_d}{2^j} \sqrt{\frac{x}{d}} + O(x^{1/4+\varepsilon} d^{1/2+\varepsilon}).$

Proof. Since $\varrho(dk) = 2f(k)$ for odd k and the sum over k is finite,

$$\mathcal{M}_d = \frac{16}{d} \int_{\mathbb{R}} F(T(t)/x) G(2^{-(v-2)/2} \sqrt{T(t)}) dt, \quad G(y) := \sum_{k \geq 1} \frac{\chi(k) f(k)}{k} W(k/y).$$

By (5.1), partial summation and Lemma 5.3(b),

$$\mathfrak{L}_d - G(y) = - \int_{y/2}^{\infty} A_f(u) \frac{d}{du} \left(\frac{1 - W(u/y)}{u} \right) du \ll y^{-1/2+\varepsilon} d^{1+\varepsilon} \quad (y \geq 1).$$

On $t > 0$ we substitute $T = T(t)$. Since $2^{v-1}E = 2^{2j-1}$ and $2^{v-2}H = m/4$, we have $dT/dt = 2^{2j-1}t/d$ and $2^j t = \sqrt{4dT - m}$, so $dt = 2^{1-j} d \sqrt{4dT - m}$. As the integrand is even in t ,

$$\mathcal{M}_d = \frac{64}{2^j} \int_{x/2}^x \frac{F(T/x) G(\sqrt{T/2^{v-2}})}{\sqrt{4dT - m}} dT.$$

Replacing G by $\mathfrak{L}_d$ costs $\ll 2^{-j} \int_{x/2}^x x^{-1/4+\varepsilon} d^{1+\varepsilon} (dT)^{-1/2} dT \ll x^{1/4+\varepsilon} d^{1/2+\varepsilon}$. Replacing $(4dT - m)^{-1/2}$ by $(4dT)^{-1/2}$ costs $\ll |\mathfrak{L}_d| x m (dx)^{-3/2} \ll d^{3/2+\varepsilon} x^{-1/2}$, using $|\mathfrak{L}_d| \ll d^{1+\varepsilon}$. Finally $\int F(T/x) (4dT)^{-1/2} dT = c_F \sqrt{x}/(2\sqrt{d})$. $\square$

7.3. The error term.

Lemma 7.3. $\mathcal{E}_d \ll x^{1/4+\theta/2+\varepsilon} d^{3/4-3\theta/2}$. With Theorem 6.1(b) in place of Theorem 6.1(a), the proof gives $\mathcal{E}_d \ll x^{1/4+\theta/2+\varepsilon} d^{3/4-\theta}$.

Proof. Fix a smooth $\psi_0 \geq 0$ supported in $[1, 2]$ with $\sum_{i \in \mathbb{Z}} \psi_0(t/2^{i/2}) = 1$ for all $t > 0$. For instance, take a smooth $\phi \geq 0$ supported in $[1, 2]$ and positive on $(1, 2)$, and put $\psi_0 = \phi / \sum_i \phi(\cdot/2^{i/2})$. Let $\mathcal{K} = \{2^{i/2} : i \in \mathbb{Z}, 2^{i/2} \geq 1/2\}$, so that $\sum_{K \in \mathcal{K}} \psi_0(k/K) = 1$ for every integer $k \geq 1$. Writing $\chi(k) = \sum_{\kappa \in \{1,3\}} \chi(\kappa) \mathbf{1}\{k \equiv \kappa \pmod{4}\}$, we get from (7.3) and (7.4)

$$\mathcal{E}_d = 8 \sum_{K \in \mathcal{K}} \sum_{\kappa \in \{1,3\}} \chi(\kappa) \mathcal{E}(K, \kappa),$$

where

$$\mathcal{E}(K, \kappa) = \sum_{k \equiv \kappa \pmod{4}} \psi_0\left(\frac{k}{K}\right) \left(\sum_{\substack{\ell \in \mathbb{Z} \\ dk | \ell^2 + H}} \Phi_k(\ell) - \frac{\varrho(dk)}{dk} \int_{\mathbb{R}} \Phi_k(t) dt \right).$$

If $K \geq k_+$ then $\mathcal{E}(K, \kappa) = 0$, because $\Phi_k = 0$ for $k \geq k_+$. So only $O(\log x)$ values of K contribute. Put $k_- = 2^{-3-(v-2)/2} \sqrt{x}$.

Write $\mu = dk$. Then $k \equiv \kappa \pmod{4}$ if and only if $\mu \equiv \kappa d \pmod{4d}$, and $\psi_0(k/K) = \psi_0(\mu/(dK))$. Let $\psi_2(s) = F(T(Xs)/x)$. Since $T(Xs)/x = \alpha s^2 + 1 - \alpha$ with $\alpha = EX^2/(EX^2 + H) \in [\frac{1}{2}, 1]$, the function ψ_2 is supported in $[-1, 1]$, and its derivatives are bounded in terms of F .

The case $K \leq k_-$. If $\psi_0(k/K) \neq 0$ and $F(T(t)/x) \neq 0$, then $2^{(v-2)/2} k / \sqrt{T(t)} \leq 2^{(v-2)/2} \cdot 2k_- / \sqrt{x/2} = 2^{-3/2} < 1/2$. Hence $W(\cdot) = 1$ and $\Phi_k(t) = \psi_2(t/X)$. Thus $\mathcal{E}(K, \kappa)$ is the sum Ξ of (6.1), with K replaced by dK , $\psi_1 = \psi_0$, ψ_2 as above and $\Delta = 1$.

The case $k_- < K < k_+$. Let $\widetilde{W}(w) = \int_0^\infty W(y) y^{w-1} dy$ for $\operatorname{Re} w > 0$. Since W' is supported in $[\frac{1}{2}, 2]$, integration by parts gives $\widetilde{W}(w) = -w^{-1} \int W'(y) y^w dy$, and repeating it gives $|\widetilde{W}(w)| \ll_A$

$(1 + |w|)^{-A}$ for $\operatorname{Re} w = 1$. By Mellin inversion, $W(y) = \frac{1}{2\pi i} \int_{(1)} \widetilde{W}(w) y^{-w} dw$ for $y > 0$. Hence, whenever $\psi_0(k/K) \neq 0$,

$$\psi_0\left(\frac{k}{K}\right) \Phi_k(t) = \frac{1}{2\pi i} \int_{(1)} \widetilde{W}(w) \psi_{1,w}\left(\frac{k}{K}\right) \psi_{2,w}\left(\frac{t}{X}\right) dw,$$

where

$$\psi_{1,w}(s) = \psi_0(s) s^{-w}, \quad \psi_{2,w}(s) = \psi_2(s) (\alpha s^2 + 1 - \alpha)^{w/2} \left(\frac{\sqrt{x}}{2^{(v-2)/2} K} \right)^w.$$

For $\operatorname{Re} w = 1$ we have $|(\sqrt{x}/(2^{(v-2)/2} K))^w| = \sqrt{x}/(2^{(v-2)/2} K) \leq 8$, and $\alpha s^2 + 1 - \alpha \in [\frac{1}{2}, 1]$ on the support of ψ_2 . So $\|\psi_{i,w}^{(\nu)}\|_\infty \ll_\nu (1 + |w|)^\nu$ for $i = 1, 2$. We insert this into $\mathcal{E}(K, \kappa)$, interchange the finite sums with the integral, and use $\int_{\mathbb{R}} \psi_{2,w}(t/X) dt = X \int_{\mathbb{R}} \psi_{2,w}$. This gives

$$\mathcal{E}(K, \kappa) = \frac{1}{2\pi i} \int_{(1)} \widetilde{W}(w) \Xi_w dw,$$

where Ξ_w is the sum Ξ of (6.1) with K replaced by dK , $\psi_1 = \psi_{1,w}$, $\psi_2 = \psi_{2,w}$ and $\Delta = 1 + |w|$.

Application of Theorem 6.1. The hypotheses (H) hold, with K replaced by dK . By §4, H is odd, $d \nmid H$, and N is neither a square nor three times a square. We have $X \geq N^{1/2}$, and $1 \leq dK \leq dk_+ \leq qX$ by (7.2). Moreover $q \ll d$, $H \asymp d^2$, and by (7.2) and (7.1)

$$X^{1/2} H^{1/4} \asymp x^{1/4} d^{3/4}, \quad \frac{X}{dH^{1/2}} \asymp x^{1/2} d^{-3/2} \geq 1, \quad \frac{dK}{dH^{1/2}} \leq \frac{k_+}{H^{1/2}} \ll \frac{x^{1/2}}{d}.$$

Theorem 6.1(a) with η small in terms of ε therefore gives, in both cases,

$$\Xi_w \ll (1 + |w|)^C x^{1/4 + \theta/2 + \varepsilon/2} d^{3/4 - 3\theta/2}$$

for some constant C , and the same bound for Ξ . Theorem 6.1(b) gives the same bounds with $d^{3/4 - \theta}$ in place of $d^{3/4 - 3\theta/2}$. Integrating against $|\widetilde{W}(w)|$ and summing over the $O(\log x)$ values of K and the two values of κ proves the lemma. $\square$

Proposition 7.1 follows from Lemmas 7.2 and 7.3.

Proof of Theorem 4.3. Let $\varpi < \varpi_\theta$ and let $d \leq x^\varpi$ be a good prime. Since $\varpi_\theta \leq 1/5$, (7.1) holds. By Lemma 5.3(d), the main term in Proposition 7.1 is $\geq c(\varepsilon) x^{1/2} d^{-1/2 - \varepsilon}$. The error terms are smaller by a factor

$$\ll x^{-1/4 + \varepsilon} d^{1 + 2\varepsilon} + x^{-1/4 + \theta/2 + \varepsilon} d^{5/4 - 3\theta/2 + \varepsilon} \leq x^{-1/4 + \varepsilon + \varpi(1 + 2\varepsilon)} + x^{-1/4 + \theta/2 + \varepsilon + \varpi(5/4 - 3\theta/2 + \varepsilon)}.$$

Both exponents are negative once ε is small enough in terms of ϖ and θ : the first because $\varpi < 1/4$, and the second because $\varpi(5/4 - 3\theta/2) < 1/4 - \theta/2$. With Theorem 6.1(b), the second factor becomes $x^{-1/4 + \theta/2 + \varepsilon} d^{5/4 - \theta + \varepsilon}$, and its exponent is negative for $\varpi < \varpi'_\theta \leq 1/5$, because then $\varpi(5/4 - \theta) < 1/4 - \theta/2$. $\square$

8. HEEGNER POINTS

Throughout Sections 8–12 we assume (H) and write $\Gamma = \Gamma_0(q)$. We fix a set $\mathcal{T}_q$ of representatives of the cosets $\Gamma \backslash \operatorname{SL}_2(\mathbb{Z})$.

8.1. Symmetric matrices and Heegner points. For an integer $N \geq 1$ let $\mathcal{Q}_N$ be the set of integral symmetric matrices $\mathbf{g} = \begin{pmatrix} \mathbf{a} & \mathbf{b} \\ \mathbf{b} & \mathbf{c} \end{pmatrix}$ with $\mathbf{a}, \mathbf{c} > 0$ and $\mathbf{a}\mathbf{c} - \mathbf{b}^2 = N$. The group $\mathrm{SL}_2(\mathbb{Z})$ acts on $\mathcal{Q}_N$ by $\gamma \diamond \mathbf{g} = \gamma \mathbf{g} \gamma^t$. To $\mathbf{g} \in \mathcal{Q}_N$ we attach the point

$$z(\mathbf{g}) = \frac{\mathbf{b} + i\sqrt{N}}{\mathbf{c}} \in \mathbb{H},$$

which determines $\mathbf{g}$. Every positive definite real symmetric matrix of determinant 1 has the form hh^t with $h \in \mathrm{SL}_2(\mathbb{R})$, and $h \cdot i$ depends only on hh^t : if $h = \begin{pmatrix} y^{1/2} & xy^{-1/2} \\ 0 & y^{-1/2} \end{pmatrix} k$ with $k \in \mathrm{SO}(2)$, then $hh^t = \begin{pmatrix} y+x^2/y & x/y \\ x/y & 1/y \end{pmatrix}$ and $h \cdot i = x + iy$. Hence if $\mathbf{g} = N^{1/2}hh^t$, then

$$(8.1) \quad z(\mathbf{g}) = h \cdot i, \quad \gamma \diamond \mathbf{g} = N^{1/2}(\gamma h)(\gamma h)^t, \quad z(\gamma \diamond \mathbf{g}) = \gamma \cdot z(\mathbf{g}) \quad (\gamma \in \mathrm{SL}_2(\mathbb{Z})).$$

We call $\mathbf{g}$ *reduced* if $|\mathbf{b}| \leq \mathbf{c} \leq \mathbf{a}$. This means that $z(\mathbf{g})$ lies in the closure of the standard fundamental domain for $\mathrm{SL}_2(\mathbb{Z})$, since $|z(\mathbf{g})|^2 = \mathbf{a}/\mathbf{c}$; in particular $\mathrm{Im} z(\mathbf{g}) \geq \sqrt{3}/2$. Every $\mathrm{SL}_2(\mathbb{Z})$ -orbit in $\mathcal{Q}_N$ contains a reduced element, and there are finitely many reduced elements. We fix a set Λ_N of reduced representatives of the $\mathrm{SL}_2(\mathbb{Z})$ -orbits on $\mathcal{Q}_N$.

We use the point-pair invariant $u(z, w) = |z - w|^2 / (4 \mathrm{Im} z \mathrm{Im} w)$ [18, Ch. 1]. It is invariant under the diagonal action of $\mathrm{SL}_2(\mathbb{R})$, and the hyperbolic distance satisfies $\cosh \mathrm{dist}(z, w) = 1 + 2u(z, w)$.

Lemma 8.1. *Let $N \geq 1$ be neither a square nor three times a square.*

- (a) *The stabiliser of every $\mathbf{g} \in \mathcal{Q}_N$ in $\mathrm{SL}_2(\mathbb{Z})$ is $\{\pm 1\}$.*
- (b) *Let $\mathcal{Q} \subseteq \mathcal{Q}_N$ be invariant under Γ . Then the matrices $\tau \diamond \mathbf{z}$ with $\mathbf{z} \in \Lambda_N$, $\tau \in \mathcal{T}_q$ and $\tau \diamond \mathbf{z} \in \mathcal{Q}$ form a system of representatives of the Γ -orbits on $\mathcal{Q}$.*

Proof. (a) Since $z(\mathbf{g})$ determines $\mathbf{g}$, (8.1) shows that $\gamma \diamond \mathbf{g} = \mathbf{g}$ if and only if $\gamma \cdot z(\mathbf{g}) = z(\mathbf{g})$. The stabiliser in $\mathrm{SL}_2(\mathbb{Z})$ of a point of $\mathbb{H}$ is $\{\pm 1\}$ unless the point is $\mathrm{SL}_2(\mathbb{Z})$ -equivalent to i or to $\rho = e^{2\pi i/3}$ [25, Ch. VII, Theorem 1]. Suppose that $z(\mathbf{g}) = g \cdot i$ or $z(\mathbf{g}) = g \cdot \rho$ with $g \in \mathrm{SL}_2(\mathbb{Z})$, and put $\mathbf{g}' = g^{-1} \diamond \mathbf{g} = \begin{pmatrix} \mathbf{a}' & \mathbf{b}' \\ \mathbf{b}' & \mathbf{c}' \end{pmatrix}$, so that $z(\mathbf{g}') = i$ or $z(\mathbf{g}') = \rho$ by (8.1). If $z(\mathbf{g}') = i$, then $\mathbf{b}' = 0$ and $\mathbf{c}' = \sqrt{N}$, so N is a square. If $z(\mathbf{g}') = \rho$, then $\mathbf{b}'/\mathbf{c}' = -\frac{1}{2}$ and $\sqrt{N}/\mathbf{c}' = \frac{\sqrt{3}}{2}$, so $\mathbf{c}' = -2\mathbf{b}'$ and $N = 3\mathbf{b}'^2$. Both cases are excluded.

(b) Every element of $\mathcal{Q}$ has the form $g \diamond \mathbf{z}$ with $\mathbf{z} \in \Lambda_N$ and $g \in \mathrm{SL}_2(\mathbb{Z})$. Write $g = \gamma\tau$ with $\gamma \in \Gamma$ and $\tau \in \mathcal{T}_q$. Then $g \diamond \mathbf{z} = \gamma \diamond (\tau \diamond \mathbf{z})$, and $\tau \diamond \mathbf{z} \in \mathcal{Q}$ because $\mathcal{Q}$ is Γ -invariant. So every Γ -orbit on $\mathcal{Q}$ contains one of the matrices in question. Suppose that $\tau' \diamond \mathbf{z}' = \gamma \diamond (\tau \diamond \mathbf{z})$ with $\gamma \in \Gamma$. Then $\mathbf{z}'$ and $\mathbf{z}$ lie in the same $\mathrm{SL}_2(\mathbb{Z})$ -orbit, so $\mathbf{z}' = \mathbf{z}$, and $\tau'^{-1}\gamma\tau$ stabilises $\mathbf{z}$. By (a), $\tau'^{-1}\gamma\tau = \pm 1$, so $\Gamma\tau' = \Gamma\tau$, since $-1 \in \Gamma$, and therefore $\tau' = \tau$. $\square$

8.2. The sets $\mathcal{Q}^\kappa$. Under (H) put

$$\mathcal{Q}^\kappa = \{\mathbf{g} \in \mathcal{Q}_N : \mathbf{b} \equiv 0 \pmod{E}, \mathbf{c} \equiv \kappa Ed \pmod{4Ed}\}.$$

Lemma 8.2. (a) *The map*

$$(\mu, \ell) \mapsto \begin{pmatrix} (E\ell^2 + H)/\mu & E\ell \\ E\ell & E\mu \end{pmatrix}$$

is a bijection from the set of pairs $(\mu, \ell) \in \mathbb{Z}^2$ with $\mu \geq 1$, $\mu \equiv \kappa d \pmod{4d}$ and $\mu \mid E\ell^2 + H$ onto $\mathcal{Q}^\kappa$. The image $\mathbf{g}$ of (μ, ℓ) satisfies $z(\mathbf{g}) = \ell/\mu + i\sqrt{N}/(E\mu)$.

- (b) *$\mathcal{Q}^\kappa$ is invariant under Γ .*

Proof. (a) The determinant of the image is $E(E\ell^2 + H) - E^2\ell^2 = N$, and its entries satisfy the congruences that define $\mathcal{Q}^\kappa$. Conversely, for $\mathbf{g} \in \mathcal{Q}^\kappa$ write $\mathbf{b} = E\ell$ and $\mathbf{c} = E\mu$; then $\mu \equiv \kappa d \pmod{4d}$ and $\mathbf{a}\mu = E\ell^2 + H$. The formula for $z(\mathbf{g})$ is immediate.

(b) Let $\gamma = \begin{pmatrix} \gamma_1 & \gamma_2 \\ \gamma_3 & \gamma_4 \end{pmatrix} \in \Gamma$, so that $q \mid \gamma_3$ and γ_4 is odd. The lower right entry of $\gamma \diamond \mathbf{g}$ is $\gamma_3^2\mathbf{a} + 2\gamma_3\gamma_4\mathbf{b} + \gamma_4^2\mathbf{c} \equiv \gamma_4^2\mathbf{c} \pmod{q}$. Moreover $\gamma_4^2\mathbf{c} \equiv \mathbf{c} \pmod{4Ed}$, because $4 \mid \gamma_4^2 - 1$ and $Ed \mid \mathbf{c}$.

The off-diagonal entry is $\gamma_1\gamma_3\mathbf{a} + (\gamma_1\gamma_4 + \gamma_2\gamma_3)\mathbf{b} + \gamma_2\gamma_4\mathbf{c} \equiv 0 \pmod{E}$, because E divides γ_3 , $\mathbf{b}$ and $\mathbf{c}$. $\square$

By Lemmas 8.1(b) and 8.2(b), the matrices $\tau \diamond \mathbf{z}$ with $\mathbf{z} \in \Lambda_N$, $\tau \in \mathcal{T}_q$ and $\tau \diamond \mathbf{z} \in \mathcal{Q}^\kappa$ form a system $\mathbf{g}_1, \dots, \mathbf{g}_I$ of representatives of the Γ -orbits on $\mathcal{Q}^\kappa$. We put $z_i = z(\mathbf{g}_i)$. If $\mathbf{g}_i = \tau \diamond \mathbf{z}$, then $z_i = \tau \cdot z(\mathbf{z})$ by (8.1). These are the Heegner points that occur in the proof of Theorem 6.1.

8.3. Counting.

Lemma 8.3. *Let $\mathcal{Q} \subseteq \{\mathbf{g} \in \mathcal{Q}_N : d \mid \mathbf{c}\}$. Then for every $\mathbf{g} \in \mathcal{Q}_N$*

$$(8.2) \quad \#\{\tau \in \mathcal{T}_q : \tau \diamond \mathbf{g} \in \mathcal{Q}\} \leq 12E \leq 96.$$

Proof. Let $\tau, \tau' \in \mathrm{SL}_2(\mathbb{Z})$ have bottom rows (τ_3, τ_4) and (τ'_3, τ'_4) with $(\tau'_3, \tau'_4) \equiv \lambda(\tau_3, \tau_4) \pmod{q}$ for some integer λ . Then the bottom row of $\tau'\tau^{-1}$ is $\equiv (0, \lambda) \pmod{q}$, so $\Gamma\tau' = \Gamma\tau$. Hence the map sending $\tau \in \mathcal{T}_q$ to the point $(\tau_3 : \tau_4) \in \mathbb{P}^1(\mathbb{Z}/q\mathbb{Z})$ given by its bottom row is injective. By the Chinese remainder theorem, $\mathbb{P}^1(\mathbb{Z}/q\mathbb{Z}) = \mathbb{P}^1(\mathbb{Z}/4E\mathbb{Z}) \times \mathbb{P}^1(\mathbb{F}_d)$. The lower right entry of $\tau \diamond \mathbf{g}$ is $\varphi_{\mathbf{g}}(\tau_3, \tau_4)$, where $\varphi_{\mathbf{g}}(s, t) = \mathbf{a}s^2 + 2\mathbf{b}st + \mathbf{c}t^2$. So $\tau \diamond \mathbf{g} \in \mathcal{Q}$ requires $d \mid \varphi_{\mathbf{g}}(\tau_3, \tau_4)$, which depends only on the image of $(\tau_3 : \tau_4)$ in $\mathbb{P}^1(\mathbb{F}_d)$. The form $\varphi_{\mathbf{g}}$ does not vanish identically modulo d : otherwise $d^2 \mid \mathbf{a}\mathbf{c} - \mathbf{b}^2 = N$, which is impossible because d is odd and $d \nmid H$. So it has at most two zeros in $\mathbb{P}^1(\mathbb{F}_d)$. Since $\#\mathbb{P}^1(\mathbb{Z}/2^r\mathbb{Z}) = 3 \cdot 2^{r-1}$, this gives (8.2). $\square$

Lemma 8.4. *We have $I \leq 96 \#\Lambda_N \ll_\eta N^{1/2+\eta}$, and*

$$\mathcal{P} := \sum_{i, i'=1}^I \#\{\gamma \in \Gamma : u(z_i, \gamma z_{i'}) \leq 1\} \leq 192 \sum_{\mathbf{z} \in \Lambda_N} \#\{\mathbf{w} \in \mathcal{Q}_N : u(z(\mathbf{w}), z(\mathbf{z})) \leq 1\} \ll_\eta N^{1+\eta}.$$

Proof. Reduction to $\mathrm{SL}_2(\mathbb{Z})$. The bound $I \leq 96 \#\Lambda_N$ follows from Lemma 8.3 with $\mathcal{Q} = \mathcal{Q}^\kappa$. Write $z_i = \tau_i \cdot z(\mathbf{z}_i)$ with $\mathbf{z}_i \in \Lambda_N$ and $\tau_i \in \mathcal{T}_q$. Fix i and $\mathbf{z}' \in \Lambda_N$. The map $(\gamma, \tau') \mapsto g = \tau_i^{-1}\gamma\tau'$ is a bijection from $\Gamma \times \mathcal{T}_q$ onto $\mathrm{SL}_2(\mathbb{Z})$, and $u(\tau_i z(\mathbf{z}_i), \gamma\tau' z(\mathbf{z}')) = u(z(\mathbf{z}_i), g z(\mathbf{z}'))$. Dropping the condition $\tau' \diamond \mathbf{z}' \in \mathcal{Q}^\kappa$, we get

$$\sum_{i': \mathbf{z}_{i'} = \mathbf{z}'} \#\{\gamma \in \Gamma : u(z_i, \gamma z_{i'}) \leq 1\} \leq \#\{g \in \mathrm{SL}_2(\mathbb{Z}) : u(z(\mathbf{z}_i), g z(\mathbf{z}')) \leq 1\}.$$

As $\mathbf{z}'$ runs over Λ_N and g over $\mathrm{SL}_2(\mathbb{Z})$, the matrix $\mathbf{w} = g \diamond \mathbf{z}'$ runs over $\mathcal{Q}_N$, taking each value twice by Lemma 8.1(a), and $z(\mathbf{w}) = g z(\mathbf{z}')$ by (8.1). Hence the sum over all i' is at most $2 \#\{\mathbf{w} \in \mathcal{Q}_N : u(z(\mathbf{w}), z(\mathbf{z}_i)) \leq 1\}$. Each $\mathbf{z} \in \Lambda_N$ occurs as $\mathbf{z}_i$ for at most 96 values of i , by Lemma 8.3. Summing over i gives the first inequality for $\mathcal{P}$.

Counting. Let $\varrho_N(\mathbf{c}) = \#\{\nu \bmod \mathbf{c} : \nu^2 + N \equiv 0 \pmod{\mathbf{c}}\}$. As in Lemma 5.2(b), $\varrho_N(\ell^i) \leq 2\ell^{\min(i, v_\ell(N))/2}$ for odd primes ℓ . Similarly $\varrho_N(2^i) \leq 4 \cdot 2^{\min(i, v_2(N))/2}$. Indeed, for $i \leq v_2(N)$ the solutions are the multiples of $2^{\lceil i/2 \rceil}$. For $i > v_2(N) =: \alpha$ there are no solutions unless α is even, and then a solution has the form $\nu = 2^{\alpha/2}\nu_1$, where ν_1 is taken modulo $2^{i-\alpha/2}$ and $\nu_1^2 \equiv -N2^{-\alpha} \pmod{2^{i-\alpha}}$; there are at most 4 such ν_1 modulo $2^{i-\alpha}$, each with $2^{\alpha/2}$ lifts. Hence $\varrho_N(\mathbf{c}) \leq 4 \cdot 2^{\omega(\mathbf{c})} \gcd(\mathbf{c}, N)^{1/2}$, and since $2^{\omega(n_1\mathbf{c}')} \leq \tau(n_1)\tau(\mathbf{c}')$,

$$\sum_{\mathbf{c} \leq y} \varrho_N(\mathbf{c}) \leq 4 \sum_{n_1 \mid N} n_1^{1/2} \tau(n_1) \sum_{\mathbf{c}' \leq y/n_1} \tau(\mathbf{c}') \ll y \log(2y) \tau(N)^2 \ll_\eta y N^\eta \log(2y).$$

A reduced $\mathbf{z} = \begin{pmatrix} \mathbf{a} & \mathbf{b} \\ \mathbf{b} & \mathbf{c} \end{pmatrix}$ satisfies $\mathbf{c}^2 \leq \mathbf{a}\mathbf{c} = N + \mathbf{b}^2 \leq N + \mathbf{c}^2/4$, so $\mathbf{c} \leq 2\sqrt{N/3}$, and $\mathbf{b}$ lies in an interval of length $\mathbf{c}$ and satisfies $\mathbf{b}^2 \equiv -N \pmod{\mathbf{c}}$; hence $\#\Lambda_N \leq 2 \sum_{\mathbf{c} \leq 2\sqrt{N}} \varrho_N(\mathbf{c}) \ll N^{1/2+2\eta}$.

Now fix $\mathbf{z} \in \Lambda_N$ with $z(\mathbf{z}) = x_1 + iy_1$, so that $y_1 \geq \sqrt{3}/2$, and let $\mathbf{w} = \begin{pmatrix} \mathbf{a} & \mathbf{b} \\ \mathbf{b} & \mathbf{c} \end{pmatrix} \in \mathcal{Q}_N$ with $w = z(\mathbf{w}) = x_w + iy_w$ and $u(w, z(\mathbf{z})) \leq 1$, that is, $|w - z(\mathbf{z})|^2 \leq 4y_w y_1$. Then $(y_w - y_1)^2 \leq 4y_w y_1$, so $y_w/y_1 \in [3 - 2\sqrt{2}, 3 + 2\sqrt{2}] \subset [1/6, 6]$. Also $(x_w - x_1)^2 \leq 24y_1^2$. Since $y_w = \sqrt{N}/\mathbf{c}$ and $x_w = \mathbf{b}/\mathbf{c}$,

we get $\mathfrak{c} \leq 6\sqrt{N}/y_1 < 7\sqrt{N}$, and $\mathfrak{b}$ lies in an interval of length $10\mathfrak{c}y_1 \leq 60\sqrt{N}$. Given $\mathfrak{b}$ and $\mathfrak{c}$, $\mathfrak{a}$ is determined. Therefore

$$\#\{\mathfrak{w} \in \mathcal{Q}_N : u(z(\mathfrak{w}), z(\mathfrak{z})) \leq 1\} \leq \sum_{\mathfrak{c} \leq 7\sqrt{N}} \left(\frac{60\sqrt{N}}{\mathfrak{c}} + 1 \right) \varrho_N(\mathfrak{c}) \ll_{\eta} N^{1/2+2\eta},$$

by partial summation. Together with the bound for $\#\Lambda_N$, this proves the lemma after renaming η . $\square$

9. POISSON SUMMATION AND POINCARÉ SERIES AT HEEGNER POINTS

We write $\Gamma_{\infty} = \{\pm \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} : n \in \mathbb{Z}\}$, and we put

$$(9.1) \quad h_0 = (qX)^{\eta} \frac{K}{X}, \quad Y = \frac{N^{1/2}}{EK}, \quad L = \Delta + 8(qX)^{\eta}.$$

9.1. Poisson summation. For $h \in \mathbb{Z}$ and $\mu \geq 1$ let

$$S(h; \mu) = \sum_{\substack{\nu \bmod \mu \\ E\nu^2 + H \equiv 0 \pmod{\mu}}} e\left(\frac{h\nu}{\mu}\right), \quad W_h = W_h(\psi_2) = \sum_{\mu \equiv \kappa d \pmod{4d}} \psi_1\left(\frac{\mu}{K}\right) \frac{X}{\mu} \hat{\psi}_2\left(\frac{hX}{\mu}\right) S(h; \mu).$$

Let ψ_0 be as in the proof of Lemma 7.3, so that $\psi_0 \geq 0$ is smooth, supported in $[1, 2]$, and $\sum_{i \in \mathbb{Z}} \psi_0(t/2^{i/2}) = 1$ for $t > 0$. Let $\mathcal{M} = \{2^{i/2} : i \in \mathbb{Z}, i \geq -2, 2^{i/2} \leq 2h_0\}$, and for $M \in \mathcal{M}$ put

$$\Xi_M(\psi_2) = \sum_{h \geq 1} \psi_0\left(\frac{h}{M}\right) W_h(\psi_2).$$

Lemma 9.1. *Let $\psi_2^-(t) = \psi_2(-t)$, and let $k_{\eta} = \lceil 4/\eta \rceil + 2$.*

- (a) $\Xi = \sum_{h \neq 0} W_h$, and the series converges absolutely.
- (b) $S(-h; \mu) = S(h; \mu)$, and $W_{-h}(\psi_2) = W_h(\psi_2^-)$.
- (c) $\sum_{|h| > h_0} |W_h| \ll \Delta^{k_{\eta}} (qX)^{-1}$.
- (d) $\Xi = \sum_{M \in \mathcal{M}} (\Xi_M(\psi_2) + \Xi_M(\psi_2^-)) + O(\Delta^{k_{\eta}} (qX)^{-1})$, and $\#\mathcal{M} \ll \log(qX)$.

Proof. (a) Splitting ℓ into residue classes modulo μ and applying Poisson summation, we get for each $\mu \geq 1$

$$\sum_{\substack{\ell \in \mathbb{Z} \\ \mu | E\ell^2 + H}} \psi_2\left(\frac{\ell}{X}\right) = \sum_{\substack{\nu \bmod \mu \\ E\nu^2 + H \equiv 0 \pmod{\mu}}} \sum_{m \in \mathbb{Z}} \psi_2\left(\frac{\nu + m\mu}{X}\right) = \frac{X}{\mu} \sum_{h \in \mathbb{Z}} \hat{\psi}_2\left(\frac{hX}{\mu}\right) S(h; \mu).$$

The term $h = 0$ is $(X/\mu) \varrho_{E,H}(\mu) \int \psi_2$, which cancels the subtracted term in (6.1). Integrating by parts twice gives $|\hat{\psi}_2(\xi)| \leq 2C_2 \Delta^2 (2\pi|\xi|)^{-2}$, and $|S(h; \mu)| \leq \mu$; this gives absolute convergence, because only $\mu \leq 2K$ occur.

(b) The map $\nu \mapsto -\nu$ permutes the roots of $E\nu^2 + H \equiv 0 \pmod{\mu}$, and $\hat{\psi}_2(-\xi) = \widehat{\psi_2^-}(\xi)$.

(c) Integrating by parts k_{η} times gives $|\hat{\psi}_2(\xi)| \leq 2C_{k_{\eta}} \Delta^{k_{\eta}} (2\pi|\xi|)^{-k_{\eta}}$. If $|h| > h_0$ and $\mu \leq 2K$, then $|hX/\mu| \geq (qX)^{\eta}/2$, so $|\hat{\psi}_2(hX/\mu)| \ll \Delta^{k_{\eta}} (qX)^{-\eta(k_{\eta}-2)} (\mu/hX)^2$. Hence

$$\sum_{|h| > h_0} |W_h| \ll \Delta^{k_{\eta}} (qX)^{-\eta(k_{\eta}-2)} \sum_{\mu \leq 2K} X \sum_{h \neq 0} \frac{\mu^2}{h^2 X^2} \ll \Delta^{k_{\eta}} (qX)^{-\eta(k_{\eta}-2)} \frac{K^3}{X} \leq \Delta^{k_{\eta}} (qX)^{3-\eta(k_{\eta}-2)},$$

using $K \leq qX$. Since $\eta(k_{\eta} - 2) \geq 4$, this is $\ll \Delta^{k_{\eta}} (qX)^{-1}$.

(d) By (a) and (b), $\Xi = \sum_{h \geq 1} (W_h(\psi_2) + W_h(\psi_2^-))$. Let $1 \leq h \leq h_0$. If $\psi_0(h/2^{i/2}) \neq 0$, then $h/2 \leq 2^{i/2} \leq h$, so $2^{i/2} \in \mathcal{M}$. Hence $\sum_{M \in \mathcal{M}} \psi_0(h/M) = 1$. For every $h \geq 1$ this sum lies in

$[0, 1]$. The terms with $h > h_0$ are therefore covered by (c), applied to ψ_2 and to ψ_2^- . Finally $h_0 \leq (qX)^{1+\eta}$. $\square$

Since ψ_2^- satisfies the same hypotheses as ψ_2 , it suffices to bound $\Xi_M(\psi_2)$ for each $M \in \mathcal{M}$ and every ψ_2 as in (H). If $h_0 < 1$, that is, if $K < (qX)^{-\eta}X$, then every $h \neq 0$ satisfies $|h| > h_0$, and parts (a) and (c) give $\Xi \ll \Delta^{k_\eta}(qX)^{-1}$. So we may assume $h_0 \geq 1$.

9.2. The Heegner–Poincaré identity. Recall the points $z_1, \dots, z_I$ defined after Lemma 8.2.

Lemma 9.2. *Let $g : (0, \infty) \rightarrow \mathbb{C}$ be smooth with compact support, let $h \in \mathbb{Z}$, and put*

$$P_{g,h}(z) = \sum_{\gamma \in \Gamma_\infty \backslash \Gamma} g(\operatorname{Im} \gamma z) e(h \operatorname{Re} \gamma z).$$

Then $P_{g,h}$ is a smooth Γ -invariant function on $\mathbb{H}$ with compact support modulo Γ , and

$$(9.2) \quad \sum_{\mu \equiv \kappa d \pmod{4d}} g\left(\frac{\sqrt{N}}{E\mu}\right) S(h; \mu) = \sum_{i=1}^I P_{g,h}(z_i).$$

In particular, for $h \geq 1$ let

$$(9.3) \quad g_h(y) = \psi_1\left(\frac{\sqrt{N}}{EKy}\right) \frac{EXy}{\sqrt{N}} \hat{\psi}_2\left(\frac{EhXy}{\sqrt{N}}\right),$$

which is supported in $[Y/2, Y]$, and let $P_h = P_{g_h,h}$. Then $W_h = \sum_{i=1}^I P_h(z_i)$.

Proof. Put $G(z) = g(\operatorname{Im} z) e(h \operatorname{Re} z)$. If $\mathfrak{g} \in \mathcal{Q}^\kappa$ corresponds to (μ, ℓ) as in Lemma 8.2(a), then

$$G(z(\mathfrak{g})) = g\left(\frac{\sqrt{N}}{E\mu}\right) e\left(\frac{h\ell}{\mu}\right).$$

The matrix $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \diamond \mathfrak{g}$ corresponds to $(\mu, \ell + t\mu)$, and -1 acts trivially. Hence the left-hand side of (9.2) is the sum of $G(z(\mathfrak{g}))$ over $\mathfrak{g} \in \Gamma_\infty \backslash \mathcal{Q}^\kappa$. Consider the map

$$(\Gamma_\infty \gamma, i) \longmapsto \Gamma_\infty \diamond (\gamma \diamond \mathfrak{g}_i)$$

from $(\Gamma_\infty \backslash \Gamma) \times \{1, \dots, I\}$ to $\Gamma_\infty \backslash \mathcal{Q}^\kappa$. It is well defined and surjective. It is also injective: if $\gamma_\infty \diamond (\gamma \diamond \mathfrak{g}_i) = \gamma' \diamond \mathfrak{g}_{i'}$ with $\gamma_\infty \in \Gamma_\infty$, then $i = i'$ and $\gamma'^{-1} \gamma_\infty \gamma$ stabilises $\mathfrak{g}_i$, so $\gamma'^{-1} \gamma_\infty \gamma = \pm 1$ by Lemma 8.1(a), and $\Gamma_\infty \gamma' = \Gamma_\infty \gamma$ since $-1 \in \Gamma_\infty$. Since $G(z(\gamma \diamond \mathfrak{g}_i)) = G(\gamma z_i)$ by (8.1), (9.2) follows.

Let g vanish outside $[y_1, y_2] \subset (0, \infty)$. If $\gamma \in \Gamma$ has lower left entry $\gamma_3 \neq 0$, then $\operatorname{Im} \gamma z \leq 1/(\gamma_3^2 \operatorname{Im} z) \leq 1/\operatorname{Im} z$. Hence $P_{g,h}(z) = G(z) = 0$ when $\operatorname{Im} z > \max(y_2, 1/y_1)$. Let $\mathfrak{s}$ be a cusp of Γ not equivalent to ∞ , with scaling matrix $\sigma_{\mathfrak{s}} = g_{\mathfrak{s}} \begin{pmatrix} \sqrt{w_{\mathfrak{s}}} & 0 \\ 0 & 1/\sqrt{w_{\mathfrak{s}}} \end{pmatrix}$, where $g_{\mathfrak{s}} \in \operatorname{SL}_2(\mathbb{Z})$ and $w_{\mathfrak{s}} > 0$. For $\gamma \in \Gamma$ the lower left entry of $\gamma \sigma_{\mathfrak{s}}$ is $\sqrt{w_{\mathfrak{s}}}$ times a non-zero integer, because $\gamma g_{\mathfrak{s}} \infty = \gamma \mathfrak{s} \neq \infty$. So $P_{g,h}(\sigma_{\mathfrak{s}} z) = 0$ when $\operatorname{Im} z > 1/(w_{\mathfrak{s}} y_1)$. Thus $P_{g,h}$ vanishes near every cusp. The sum defining $P_{g,h}$ is locally finite, so $P_{g,h}$ is smooth. For $g = g_h$ we have $g_h(\sqrt{N}/(E\mu)) = \psi_1(\mu/K)(X/\mu) \hat{\psi}_2(hX/\mu)$, and the support statement follows from $\operatorname{supp} \psi_1 \subseteq [1, 2]$. $\square$

For $M \in \mathcal{M}$ we put $P_M = \sum_{h \geq 1} \psi_0(h/M) P_h$. This is a finite sum, and Lemmas 9.1 and 9.2 give

$$(9.4) \quad \Xi_M(\psi_2) = \sum_{i=1}^I P_M(z_i).$$

10. SPECTRAL EXPANSION AND LARGE SIEVE INEQUALITIES

In Sections 10–12 the letter T denotes a bound for spectral parameters; the numbers T of Sections 2–7 do not occur there.

10.1. Normalisations. Let $\{u_j\}$ be an orthonormal basis of the space of Maass cusp forms for Γ , with $\Delta_{\mathbb{H}}u_j = (\frac{1}{4} + t_j^2)u_j$ and $t_j \in [0, \infty) \cup i(0, \frac{1}{2})$. Here $\Delta_{\mathbb{H}} = -y^2(\partial_x^2 + \partial_y^2)$, and $L^2(\Gamma \backslash \mathbb{H})$ carries the inner product $\langle f, g \rangle = \int_{\Gamma \backslash \mathbb{H}} f \bar{g} d\mu_{\mathbb{H}}$ with $d\mu_{\mathbb{H}} = dx dy/y^2$. The forms with $t_j = i\theta_j$, $\theta_j > 0$, are called exceptional; by the admissibility of θ they satisfy $\theta_j \leq \theta$. We write $\sum_{\text{exc}}$ for a sum over them. We use the normalisation of [23, (3.15)]:

$$u_j(z) = y^{1/2} \sum_{n \neq 0} \rho_j(n) K_{it_j}(2\pi|n|y) e(nx).$$

For a cusp $\mathfrak{s}$ of Γ (with a fixed scaling matrix) the Eisenstein series $E_{\mathfrak{s}}(z, s)$ has the expansion

$$E_{\mathfrak{s}}(z, \tfrac{1}{2} + ir) = \delta_{\mathfrak{s}\infty} y^{1/2+ir} + \varphi_{\mathfrak{s}\infty}(\tfrac{1}{2} + ir) y^{1/2-ir} + y^{1/2} \sum_{n \neq 0} \rho_{\mathfrak{s}}(n, r) K_{ir}(2\pi|n|y) e(nx),$$

where

$$\rho_{\mathfrak{s}}(n, r) = \frac{2\pi^{1/2+ir}}{\Gamma(\frac{1}{2} + ir)} |n|^{ir} \varphi_{\mathfrak{s}\infty}(n; \tfrac{1}{2} + ir), \quad \varphi_{\mathfrak{s}\infty}(n; s) = \sum_c c^{-2s} S_{\mathfrak{s}\infty}(0, n; c),$$

by [18, Theorem 3.4]. Here $\varphi_{\mathfrak{s}\infty}(s)$ is the scattering coefficient, $S_{\mathfrak{s}\infty}(0, n; c) = \sum e(nd/c)$, where the sum runs over the matrices $\begin{pmatrix} * & * \\ c & d \end{pmatrix} \in \Gamma_{\infty} \backslash \sigma_{\mathfrak{s}}^{-1} \Gamma / \Gamma_{\infty}$ with lower left entry c , and the sum over c runs over the positive lower left entries of the matrices in $\sigma_{\mathfrak{s}}^{-1} \Gamma$. The series $\varphi_{\mathfrak{s}\infty}(n; s)$ is the series of [23, (3.16)] for the pair of cusps $(\mathfrak{s}, \infty)$, and, as in [23, (3.9)], the scaling matrix at ∞ is the identity. A smooth function f on $\Gamma \backslash \mathbb{H}$ with compact support and $\langle f, 1 \rangle = 0$ has the expansion

$$(10.1) \quad f(z) = \sum_j \langle f, u_j \rangle u_j(z) + \frac{1}{4\pi} \sum_{\mathfrak{s}} \int_{\mathbb{R}} \langle f, E_{\mathfrak{s}}(\cdot, \tfrac{1}{2} + ir) \rangle E_{\mathfrak{s}}(z, \tfrac{1}{2} + ir) dr,$$

which converges absolutely and uniformly on compact sets [18, Theorem 7.3].

10.2. The expansion of Ξ_M . For $h \geq 1$ and $t \in \mathbb{R} \cup i(-\frac{1}{2}, \frac{1}{2})$ let

$$(10.2) \quad \check{g}_h(t) = \int_0^{\infty} g_h(y) K_{it}(2\pi hy) y^{-3/2} dy = \frac{X}{K} Y^{-1/2} \int_{1/2}^1 w_1(s) a_h(s) K_{it}(2\pi h Y s) ds,$$

where

$$w_1(s) = s^{-1/2} \psi_1(1/s), \quad a_h(s) = \hat{\psi}_2\left(\frac{hXs}{K}\right).$$

The second expression follows from (9.3) by the substitution $y = Ys$. For $M \in \mathcal{M}$ define

$$U_j = \sum_{i=1}^I u_j(z_i), \quad U_{\mathfrak{s}}(r) = \sum_{i=1}^I E_{\mathfrak{s}}(z_i, \tfrac{1}{2} + ir),$$

$$B_j = \sum_{h \geq 1} \psi_0\left(\frac{h}{M}\right) \overline{\rho_j(h)} \check{g}_h(t_j), \quad B_{\mathfrak{s}}(r) = \sum_{h \geq 1} \psi_0\left(\frac{h}{M}\right) \overline{\rho_{\mathfrak{s}}(h, r)} \check{g}_h(r).$$

Lemma 10.1. For $M \in \mathcal{M}$,

$$\Xi_M(\psi_2) = \sum_j B_j U_j + \frac{1}{4\pi} \sum_{\mathfrak{s}} \int_{\mathbb{R}} B_{\mathfrak{s}}(r) U_{\mathfrak{s}}(r) dr,$$

where the sum and the integrals converge absolutely.

Proof. By Lemma 9.2, P_M is smooth with compact support modulo Γ . Unfolding the sum over $\Gamma_\infty \backslash \Gamma$ gives $\langle P_h, 1 \rangle = \int_0^\infty \int_0^1 g_h(y) e(hx) dx dy / y^2 = 0$ and

$$\langle P_h, u_j \rangle = \int_0^\infty \int_0^1 g_h(y) e(hx) \overline{u_j(z)} \frac{dx dy}{y^2} = \overline{\rho_j(h)} \int_0^\infty g_h(y) K_{it_j}(2\pi hy) y^{-3/2} dy = \overline{\rho_j(h)} \check{g}_h(t_j),$$

because $K_{it}(y)$ is real for $t \in \mathbb{R} \cup i\mathbb{R}$. Similarly $\langle P_h, E_s(\cdot, \frac{1}{2} + ir) \rangle = \overline{\rho_s(h, r)} \check{g}_h(r)$; the constant term of E_s does not contribute because $h \neq 0$. Now apply (10.1) to P_M at the points z_i and use (9.4). $\square$

10.3. The inequalities of Deshouillers–Iwaniec and Pascadi. In the normalisation above, Pascadi's exceptional parameter is $\sqrt{1 - 4\lambda_j} = 2\theta_j$, where $\lambda_j = \frac{1}{4} - \theta_j^2$, so his weights X^{θ_j} become our $\mathcal{Y}^{2\theta_j}$. The cusp ∞ of $\Gamma_0(q)$, with the identity as scaling matrix, has $\mu(\infty) = q^{-1}$ in the notation of [23]. In [23] a hypothesis $\mathcal{Y} \ll Z$ means $\mathcal{Y} \leq cZ$ for a constant c . Since $\theta_j < \frac{1}{2}$, the weights change by at most a constant factor when $\mathcal{Y}$ is multiplied by a constant, so we may state the ranges below with $c = 1$. In this subsection and the next, $M \geq \frac{1}{2}$ is the length of the sequences. For exceptional j we have $\cosh(\pi t_j) = \cos(\pi \theta_j) \in (0, 1]$.

Lemma 10.2 ([4, Theorem 2]; see [23, Lemma H]). *Let $T \geq 1$, $M \geq \frac{1}{2}$ and $(a_n)_{n \sim M}$ complex. Then*

$$\sum_{|t_j| \leq T} \frac{1}{\cosh(\pi t_j)} \left| \sum_{n \sim M} a_n \rho_j(n) \right|^2 + \sum_s \int_{-T}^T \left| \sum_{n \sim M} a_n \rho_s(n, r) \right|^2 \frac{dr}{\cosh(\pi r)} \ll_\eta \left(T^2 + \frac{M^{1+\eta}}{q} \right) \|a\|_2^2.$$

The first sum includes the exceptional forms.

Proof. This is [4, Theorem 2], in the form and normalisation of [23, Lemma H], for the cusp ∞ ; as noted in [23], the sum over the Maass forms includes the exceptional spectrum. For the Eisenstein part note that $|\Gamma(\frac{1}{2} + ir)|^2 = \pi / \cosh(\pi r)$, so $|\sum_n a_n \rho_s(n, r)|^2 / \cosh(\pi r) = 4 |\sum_n a_n n^{ir} \varphi_{s\infty}(n; \frac{1}{2} + ir)|^2$. $\square$

Lemma 10.3 ([4, Theorem 5]; see [23, Theorem A]). *Let $M \geq \frac{1}{2}$, $(a_n)_{n \sim M}$ complex and $1 \leq \mathcal{Y} \leq \max(1, q/M)$. Then*

$$\sum_{\text{exc}} \mathcal{Y}^{2\theta_j} \left| \sum_{n \sim M} a_n \rho_j(n) \right|^2 \ll_\eta (qM)^\eta \left(1 + \frac{M}{q} \right) \|a\|_2^2.$$

Lemma 10.3 is used only in the proof of Theorem 6.1(b), and the next two lemmas only in the proof of Theorem 6.1(a).

Lemma 10.4 ([23, Theorem 2]). *Let $M \geq \frac{1}{2}$, $\alpha \in \mathbb{R}$ and $1 \leq \mathcal{Y} \leq \max(M, q)/(1 + M\|\alpha\|)$. Then*

$$\sum_{\text{exc}} \mathcal{Y}^{2\theta_j} \left| \sum_{n \sim M} e(n\alpha) \rho_j(n) \right|^2 \ll_\eta (qM)^\eta \left(1 + \frac{M}{q} \right) M.$$

Proof. This is [23, Theorem 2] for the cusp ∞ , with the dilation parameter of [23] (called a there) equal to 1. In our notation the range there is

$$\mathcal{Y} \ll \frac{\max(M, q)}{\min_{t \in \mathbb{Z}_+} (t + M\|\alpha\|)},$$

and the minimum is at most $1 + M\|\alpha\|$ (take $t = 1$). $\square$

Pascadi remarks after [23, Theorem 2] that the phases $e(n\alpha)$ may be multiplied by a fixed smooth weight $\Phi(n/M)$. The weights that occur below depend on parameters, so we need a quantitative version, which we derive by Fourier inversion.

Lemma 10.5. *Let $M \geq \frac{1}{2}$, $\mathcal{Y} \geq 1$, $L_G \geq 1$ and $C_G > 0$, and let $G : \mathbb{R} \rightarrow \mathbb{C}$ be smooth, supported in $[1, 2]$, with $\|G^{(k)}\|_\infty \leq C_G L_G^k$ for $0 \leq k \leq 3$. Then*

$$\sum_{\text{exc}} \mathcal{Y}^{2\theta_j} \left| \sum_n G\left(\frac{n}{M}\right) \rho_j(n) \right|^2 \ll_\eta C_G^2 L_G^3 (qM)^\eta \left(1 + \frac{\mathcal{Y}}{\max(M, q)}\right)^{2\theta} \left(1 + \frac{M}{q}\right) M.$$

Proof. Since G is smooth and supported in $[1, 2]$, $G(n/M) \neq 0$ implies $n \sim M$, and $G(n/M) = \int_{\mathbb{R}} \hat{G}(\xi) e(n\xi/M) d\xi$. By Cauchy–Schwarz,

$$\left| \sum_n G\left(\frac{n}{M}\right) \rho_j(n) \right|^2 \leq \|\hat{G}\|_1 \int_{\mathbb{R}} |\hat{G}(\xi)| \left| \sum_{n \sim M} e\left(\frac{n\xi}{M}\right) \rho_j(n) \right|^2 d\xi.$$

Fix ξ and put $R_\xi = \max(M, q)/(1 + |\xi|)$. Since $M\|\xi/M\| \leq |\xi|$, Lemma 10.4 applies with $\alpha = \xi/M$ for every $\mathcal{Y}' \in [1, R_\xi]$ in place of $\mathcal{Y}$. If $\mathcal{Y} \leq R_\xi$, we apply it with $\mathcal{Y}' = \mathcal{Y}$. If $1 \leq R_\xi < \mathcal{Y}$, we use $\mathcal{Y}^{2\theta_j} \leq R_\xi^{2\theta_j} (\mathcal{Y}/R_\xi)^{2\theta}$ and apply it with $\mathcal{Y}' = R_\xi$. If $R_\xi < 1$, we use $\mathcal{Y}^{2\theta_j} \leq (\mathcal{Y}/R_\xi)^{2\theta}$ and Lemma 10.2 with $T = 1$, recalling that $\cosh(\pi t_j) \leq 1$ for exceptional j . In all cases

$$\sum_{\text{exc}} \mathcal{Y}^{2\theta_j} \left| \sum_{n \sim M} e\left(\frac{n\xi}{M}\right) \rho_j(n) \right|^2 \ll (qM)^\eta \left(1 + \frac{\mathcal{Y}}{R_\xi}\right)^{2\theta} \left(1 + \frac{M}{q}\right) M,$$

and

$$\left(1 + \frac{\mathcal{Y}}{R_\xi}\right)^{2\theta} \leq (1 + |\xi|) \left(1 + \frac{\mathcal{Y}}{\max(M, q)}\right)^{2\theta},$$

because $2\theta < 1$. Finally $|\hat{G}(\xi)| \leq C_G \min(1, L_G^3 (2\pi|\xi|)^{-3})$, so $\|\hat{G}\|_1 \ll C_G L_G$ and $\int_{\mathbb{R}} |\hat{G}(\xi)| (1 + |\xi|) d\xi \ll C_G L_G^2$. $\square$

10.4. Removing a twist. The next lemma removes a smooth twist that depends on j ; it is partial summation.

Lemma 10.6. *Let $M \geq \frac{1}{2}$, let $(\xi_j(n))_{n \sim M}$ be complex numbers and $\omega_j \geq 0$ for j in a countable set, and let $\varphi_j : [M, 2M] \rightarrow \mathbb{C}$ be continuously differentiable with $|\varphi_j| \leq D_0$ and $|\varphi'_j| \leq D_1$ for all j . Then*

$$\sum_j \omega_j \left| \sum_{n \sim M} \xi_j(n) \varphi_j(n) \right|^2 \leq 2(D_0^2 + M^2 D_1^2) \sup_{M \leq y \leq 2M} \sum_j \omega_j \left| \sum_{M < n \leq y} \xi_j(n) \right|^2.$$

Proof. Let $S_j(y) = \sum_{M < n \leq y} \xi_j(n)$. Partial summation gives

$$\sum_{n \sim M} \xi_j(n) \varphi_j(n) = \varphi_j(2M) S_j(2M) - \int_M^{2M} \varphi'_j(y) S_j(y) dy.$$

By Cauchy–Schwarz, the square of its absolute value is at most

$$2D_0^2 |S_j(2M)|^2 + 2MD_1^2 \int_M^{2M} |S_j(y)|^2 dy.$$

Multiply by ω_j and sum over j . $\square$

10.5. The Heegner side. Recall the number $\mathcal{P}$ of pairs of Heegner points at bounded distance from Lemma 8.4.

Lemma 10.7. *For $T \geq 1$,*

$$\sum_{t_j \in [0, T]} |U_j|^2 + \sum_{\text{exc}} |U_j|^2 + \frac{1}{4\pi} \sum_{\mathfrak{s}} \int_{-T}^T |U_{\mathfrak{s}}(r)|^2 dr \leq \frac{256}{\pi} T^2 \mathcal{P}.$$

In particular, by Lemma 8.4, the left-hand side is $\ll_\eta T^2 N^{1+\eta}$.

Proof. Let $\beta_0 = 1/(64T^2)$. Fix a smooth $\phi_0 : [0, \infty) \rightarrow [0, 1]$ with $\phi_0 = 1$ on $[0, \frac{1}{2}]$ and $\phi_0 = 0$ on $[1, \infty)$, and let $k_0(z, w) = \phi_0(u(z, w)/\beta_0)$. Let $k = k_0 * k_0$, that is, $k(z, w) = \int_{\mathbb{H}} k_0(z, z') k_0(z', w) d\mu_{\mathbb{H}}(z')$. This is a smooth point-pair invariant with compact support. Let

$$\tilde{k}_0(t) = \int_{\mathbb{H}} k_0(z, i) (\operatorname{Im} z)^{1/2+it} d\mu_{\mathbb{H}}(z)$$

be the Selberg transform of k_0 [18, Ch. 1]. The integral operator with kernel k_0 acts on every eigenfunction of $\Delta_{\mathbb{H}}$ with eigenvalue $\frac{1}{4} + t^2$ as multiplication by $\tilde{k}_0(t)$. The operator with kernel k is its square, so the Selberg transform of k is $\tilde{k} = \tilde{k}_0^2$. The function $\tilde{k}_0$ is even. Since $\overline{\tilde{k}_0(t)} = \tilde{k}_0(-\bar{t})$, it is real on $\mathbb{R} \cup i\mathbb{R}$, so $\tilde{k} \geq 0$ there.

The disc $\{z' : u(z, z') \leq U\}$ has hyperbolic area $4\pi U$, since $u(z, z') = \sinh^2(\operatorname{dist}(z, z')/2)$. Hence $0 \leq k \leq 4\pi\beta_0$. If $k(z, w) \neq 0$, there is z' with $u(z, z'), u(z', w) \leq \beta_0$. Let $R_0 > 0$ be defined by $\cosh R_0 = 1 + 2\beta_0$. Then $\operatorname{dist}(z, w) \leq 2R_0$, and $u(z, w) \leq \sinh^2 R_0 = 4\beta_0 + 4\beta_0^2 \leq 1$.

Next we bound $\tilde{k}_0$ from below. On the support of $k_0(\cdot, i)$ we have $|\log \operatorname{Im} z| \leq \operatorname{dist}(z, i) \leq R_0 \leq 2\sqrt{\beta_0} = 1/(4T)$, because $\cosh R - 1 \geq R^2/2$. For real $|t| \leq T$ this gives $|(\frac{1}{2} + it) \log \operatorname{Im} z| \leq 3/8$, hence $\operatorname{Re}(\operatorname{Im} z)^{1/2+it} \geq e^{-3/8} \cos(3/8) > \frac{1}{2}$. For $t = i\theta_j$ the integrand is $(\operatorname{Im} z)^{1/2-\theta_j} k_0(z, i) \geq \frac{1}{2} k_0(z, i)$. In both cases $\tilde{k}_0(t) \geq \frac{1}{2} \int_{\mathbb{H}} k_0(z, i) d\mu_{\mathbb{H}}(z) \geq \frac{1}{2} \cdot 4\pi \cdot \frac{\beta_0}{2} = \pi\beta_0$.

The pre-trace formula [18, Theorem 7.4], applied at the points z_i and summed, gives

$$\sum_j \tilde{k}(t_j) |U_j|^2 + \frac{1}{4\pi} \sum_s \int_{\mathbb{R}} \tilde{k}(r) |U_s(r)|^2 dr \leq \sum_{i,i'=1}^I \sum_{\gamma \in \Gamma} k(z_i, \gamma z_{i'}),$$

where the sum over j now also includes the constant eigenfunction, with a non-negative contribution. The automorphic kernel in [18, Theorem 7.4] is a sum over $\Gamma/\{\pm 1\}$, while the sum on the right is over the matrices $\gamma \in \Gamma$, as in the definition of $\mathcal{P}$. Since $-1 \in \Gamma$ and $k \geq 0$, the right-hand side is twice the automorphic kernel; this is why we write an inequality. All terms on the left are non-negative, and the right-hand side is at most $4\pi\beta_0\mathcal{P}$. Dropping the terms with $|t_j| > T$, $|r| > T$ and the constant, we get $(\pi\beta_0)^2 \cdot (\text{left-hand side of the lemma}) \leq 4\pi\beta_0\mathcal{P}$, which is the assertion. $\square$

11. THE BESSEL TRANSFORM

Lemma 11.1. *For $y > 0$ and $\nu \in \mathbb{C}$,*

$$K_{\nu}(y) = \frac{1}{2} \int_{\mathbb{R}} \exp(-y \cosh \sigma + \nu \sigma) d\sigma.$$

If $|\operatorname{Re} \nu| \leq \frac{1}{2}$, $V \geq 1$ and $y \geq y_0 > 0$, then the integral over $|\sigma| > \log V$ contributes at most $(2/y_0)e^{-y_0 V/2}$ in absolute value.

Proof. The formula is $K_{\nu}(y) = \int_0^{\infty} e^{-y \cosh \sigma} \cosh(\nu \sigma) d\sigma$ [29, §6.22], written as an integral over $\mathbb{R}$. For $|\sigma| \geq \log V$ we have $\cosh \sigma \geq e^{|\sigma|}/2$ and $|e^{\nu \sigma}| \leq e^{|\sigma|/2} \leq V^{-1/2} e^{|\sigma|}$. Hence each of the ranges $\sigma \geq \log V$ and $\sigma \leq -\log V$ contributes at most

$$\frac{V^{-1/2}}{2} \int_{\log V}^{\infty} e^{\sigma} \exp\left(-\frac{y}{2} e^{\sigma}\right) d\sigma = \frac{V^{-1/2}}{y} e^{-yV/2} \leq \frac{1}{y_0} e^{-y_0 V/2}. \quad \square$$

Lemma 11.2. *Let $t \in \mathbb{R} \setminus \{0\}$ and $y > 0$. Put $G_t(w) = 2^{w-2} \Gamma(\frac{w+it}{2}) \Gamma(\frac{w-it}{2})$. Then*

$$K_{it}(y) = \frac{1}{2} \Gamma(it) \left(\frac{y}{2}\right)^{-it} + \frac{1}{2} \Gamma(-it) \left(\frac{y}{2}\right)^{it} + \frac{1}{2\pi i} \int_{(-3/2)} G_t(w) y^{-w} dw,$$

where the integral converges absolutely. There is an absolute constant C such that

$$|G_t(-\frac{3}{2} + i\xi)| \leq C e^{-\pi|t|/2} (1 + |\xi + t|)^{-5/4} (1 + |\xi - t|)^{-5/4} \quad (\xi \in \mathbb{R}).$$

Proof. For $\operatorname{Re} w > 0$ we have $\int_0^\infty K_{it}(y) y^{w-1} dy = G_t(w)$ [11, 6.561.16]. Mellin inversion on $\operatorname{Re} w = 1$ and a shift of the contour to $\operatorname{Re} w = -\frac{3}{2}$ give the formula. The shift is justified by the exponential decay of $G_t(w)$ in $|\operatorname{Im} w|$, which follows from Stirling's formula. The only poles in $-\frac{3}{2} < \operatorname{Re} w < 1$ are at $w = \pm it$. Since $\Gamma(z)$ has residue 1 at $z = 0$, the residue of $G_t(w) y^{-w}$ at $w = \pm it$ is $2 \cdot 2^{\pm it-2} \Gamma(\pm it) y^{\mp it} = \frac{1}{2} \Gamma(\pm it) (y/2)^{\mp it}$. For the bound, Stirling's formula gives $|\Gamma(-\frac{3}{4} + i\tau)| \ll (1 + |\tau|)^{-5/4} e^{-\pi|\tau|/2}$ for all real τ . Moreover $|\xi + t| + |\xi - t| \geq 2|t|$. $\square$

For $h > 0$ and $w \in \mathbb{C}$ let

$$M_h(w) = \int_{1/2}^1 w_1(s) a_h(s) s^{-w} ds.$$

Lemma 11.3. *Let $h > 0$ with $hX/K \leq 4(qX)^\eta$. Then $\|(w_1 a_h)^{(k)}\|_\infty \ll_k L^k$ for $k \geq 0$. For $\operatorname{Re} w \leq 0$ and $k \geq 0$,*

$$|M_h(w)| \ll_k L^k (1 + |w|)^{-k},$$

and for $\operatorname{Re} w = 0$ we have $|\partial_h M_h(w)| \ll X/K$.

Proof. By the hypothesis on ψ_1 , the derivatives of w_1 satisfy $w_1^{(i)} \ll_i \Delta^i$ on $[\frac{1}{2}, 1]$. We have $a_h^{(i)}(s) = (hX/K)^i \hat{\psi}_2^{(i)}(hXs/K)$ and $|\hat{\psi}_2^{(i)}| \leq (2\pi)^i \|\psi_2\|_1 \leq 2(2\pi)^i C_0$. This gives the first assertion, since $\Delta + hX/K \leq L$. Next, $g = w_1 a_h$ is smooth and supported in $[\frac{1}{2}, 1]$. Integrating by parts k times gives

$$M_h(w) = (-1)^k \prod_{i=1}^k (i - w)^{-1} \int_{1/2}^1 g^{(k)}(s) s^{k-w} ds.$$

For $\operatorname{Re} w \leq 0$ and $i \geq 1$ we have $|i - w| \geq (1 + |w|)/\sqrt{2}$, and $|s^{k-w}| \leq 1$ for $s \leq 1$. The last assertion follows from $\partial_h a_h(s) = (Xs/K) \hat{\psi}_2'(hXs/K)$. $\square$

12. PROOF OF THEOREM 6.1

12.1. Set-up. We may assume that $0 < \eta \leq 1$, since the bound of Theorem 6.1 for $\eta = 1$ implies it for every larger η , and that $h_0 \geq 1$ (see the remark after Lemma 9.1). Fix $M \in \mathcal{M}$ and ψ_2 as in (H). If $\psi_0(h/M) \neq 0$, then $M < h < 2M$. Since $M \leq 2h_0$, for such h and for $s \in [\frac{1}{2}, 1]$ we have

$$(12.1) \quad \pi MY \leq 2\pi hYs \leq 4\pi MY, \quad \frac{1}{16qX} \leq MY \leq \frac{(qX)^\eta}{2}, \quad \frac{hX}{K} \leq 4(qX)^\eta,$$

using $X \geq \sqrt{N}$, $N \geq 1$, $4 \leq E \leq 8$, $K \leq qX$ and $M \geq \frac{1}{2}$. In particular Lemma 11.3 applies to all h that occur. Put

$$V = \frac{(qX)^\eta}{MY} \geq 1, \quad T_1 = (qX)^\eta L, \\ \mathfrak{E}_P = \left(1 + \frac{V}{\max(M, q)}\right)^\theta, \quad \mathfrak{E}_{DI} = \left(1 + \frac{V}{\max(1, q/M)}\right)^\theta.$$

Write $\check{g}_h(t) = (X/K) Y^{-1/2} \Psi_h(t)$, with $\Psi_h(t) = \int w_1(s) a_h(s) K_{it}(2\pi hYs) ds$. We split the spectrum into

$$\mathcal{J}_0 = \{j : t_j \in [0, 1]\} \cup \{j \text{ exceptional}\}, \quad \mathcal{J}_1 = \{j : t_j > 1\},$$

and the continuous spectrum into $|r| \leq 1$ and $|r| > 1$. We shall prove

$$(12.2) \quad \Xi_M(\psi_2) \ll L^{O(1)} (qXN)^{O(\eta)} \frac{X}{K} Y^{-1/2} N^{1/2} \left(\left(1 + \frac{M}{q}\right) M \right)^{1/2} \mathfrak{E} + L^2 (qX)^{-4},$$

with $\mathfrak{E} = \mathfrak{E}_P$, respectively $\mathfrak{E} = \mathfrak{E}_{DI}$ if Lemma 10.3 is used in place of Lemma 10.5. Here $L^{O(1)}$ and $(qXN)^{O(\eta)}$ denote factors L^c and $(qXN)^{c\eta}$ with absolute constants c .

Several polynomial bounds will be used to discard negligible terms. By (12.1), $X/K \leq X$, $Y^{-1/2} \leq (16qXM)^{1/2}$, $N \leq X^2$ and $M \leq 2h_0 \leq 2(qX)^{1+\eta}$. By Lemma 10.2 with a single non-zero coefficient, $\sum_{|t_j| \leq T} |\rho_j(h)|^2 / \cosh(\pi t_j) \ll T^2 + h^{1+\eta}/q$; this includes the exceptional j , for which $\cosh(\pi t_j) \leq 1$. And by Lemma 10.7, $\sum_{t_j \in [0, T]} |U_j|^2 \ll T^2 N^{1+\eta}$.

12.2. The spectrum $\mathcal{J}_0$. Let $j \in \mathcal{J}_0$. Then $|\operatorname{Re}(it_j)| \leq \theta < \frac{1}{2}$. Lemma 11.1 with $y_0 = \pi MY$ and our V , together with (12.1), gives

$$\Psi_h(t_j) = \frac{1}{2} \int_{-\log V}^{\log V} e^{it_j \sigma} c_h(\sigma) d\sigma + O(qX e^{-\pi(qX)^\eta/2}),$$

where

$$c_h(\sigma) = \int_{1/2}^1 w_1(s) a_h(s) \exp(-2\pi h Y s \cosh \sigma) ds.$$

Hence $B_j = (X/K)Y^{-1/2}(\frac{1}{2} \int_{-\log V}^{\log V} e^{it_j \sigma} C_j(\sigma) d\sigma + R_j)$, where

$$C_j(\sigma) = \sum_h \psi_0\left(\frac{h}{M}\right) \overline{\rho_j(h)} c_h(\sigma), \quad |R_j| \ll qX e^{-\pi(qX)^\eta/2} \sum_{h \sim M} |\rho_j(h)|.$$

By Cauchy–Schwarz with respect to $d\sigma$ on $[-\log V, \log V]$,

$$(12.3) \quad |B_j|^2 \leq 2 \frac{X^2}{K^2} Y^{-1} \left(\frac{\log V}{2} \int_{-\log V}^{\log V} |e^{it_j \sigma}|^2 |C_j(\sigma)|^2 d\sigma + |R_j|^2 \right).$$

For real t_j we have $|e^{it_j \sigma}| = 1$. For exceptional j we have $|e^{it_j \sigma}|^2 = e^{-2\theta_j \sigma} \leq V^{2\theta_j}$ for $|\sigma| \leq \log V$. Since $|c_h(\sigma)| \leq \|w_1\|_1 \|\hat{\psi}_2\|_\infty \ll 1$, Lemma 10.2 gives

$$\sum_{t_j \in [0, 1]} |C_j(\sigma)|^2 \leq \cosh(\pi) \sum_{t_j \in [0, 1]} \frac{|C_j(\sigma)|^2}{\cosh(\pi t_j)} \ll \left(1 + \frac{M^{1+\eta}}{q}\right) M.$$

For the exceptional spectrum, note that $C_j(\sigma) = \overline{\sum_n G_\sigma(n/M) \rho_j(n)}$ with

$$G_\sigma(y) = \psi_0(y) \int_{1/2}^1 \overline{w_1(s)} \hat{\psi}_2\left(\frac{y M X s}{K}\right) \exp(-2\pi y M Y s \cosh \sigma) ds.$$

By (12.1) and the bound $\lambda^k e^{-y\lambda} \leq k^k$ for $y \geq 1$, $\lambda \geq 0$, we have $\|G_\sigma^{(k)}\|_\infty \ll_k L^k$ uniformly in $\sigma \in \mathbb{R}$. So Lemma 10.5 with $\mathcal{Y} = V$, $L_G = L$ and $C_G \ll 1$ gives

$$\sum_{\text{exc}} V^{2\theta_j} |C_j(\sigma)|^2 \ll L^3 (qM)^\eta \mathfrak{E}_P^2 \left(1 + \frac{M}{q}\right) M.$$

If instead we use Lemma 10.3, we argue as follows. If $V \leq \max(1, q/M)$, we apply it directly. Otherwise we bound $V^{2\theta_j} \leq \max(1, q/M)^{2\theta_j} (V/\max(1, q/M))^{2\theta}$ and apply it with $\mathcal{Y} = \max(1, q/M)$. This gives the same bound with $\mathfrak{E}_{DI}$ in place of $\mathfrak{E}_P$, and without the factor L^3 . The terms R_j are negligible: by the polynomial bounds above, $\sum_{j \in \mathcal{J}_0} |R_j|^2 \ll (qX)^{-20}$. Since $\log V \ll \log(qX)$, we obtain from (12.3)

$$\sum_{j \in \mathcal{J}_0} |B_j|^2 \ll \frac{X^2}{K^2} Y^{-1} (\log qX)^2 L^3 (qM)^\eta \mathfrak{E}^2 \left(1 + \frac{M^{1+\eta}}{q}\right) M,$$

since $\mathfrak{E} \geq 1$. Lemma 10.7 with $T = 1$ gives $\sum_{j \in \mathcal{J}_0} |U_j|^2 \ll N^{1+\eta}$. By the Cauchy–Schwarz inequality, the sum $\sum_{j \in \mathcal{J}_0} |B_j U_j|$ is therefore bounded by the right-hand side of (12.2).

12.3. The spectrum $\mathcal{J}_1$: the main terms of the Bessel functions. Let $j \in \mathcal{J}_1$. Lemma 11.2 with $y = 2\pi hYs$, integrated against $w_1(s)a_h(s)$, gives

$$\begin{aligned} \Psi_h(t_j) &= \frac{1}{2}\Gamma(it_j)(\pi hY)^{-it_j}M_h(it_j) + \frac{1}{2}\Gamma(-it_j)(\pi hY)^{it_j}M_h(-it_j) \\ &\quad + \frac{1}{2\pi i} \int_{(-3/2)} G_{t_j}(w)(2\pi hY)^{-w}M_h(w) dw. \end{aligned}$$

The interchange of integrals is justified by absolute convergence. Accordingly $B_j = B_j^+ + B_j^- + B_j^R$, with

$$B_j^\pm = \frac{X}{K}Y^{-1/2} \frac{1}{2}\Gamma(\pm it_j) \sum_h \psi_0\left(\frac{h}{M}\right) \overline{\rho_j(h)} (\pi hY)^{\mp it_j} M_h(\pm it_j).$$

For $|t| \geq 1$ we have $|\Gamma(it)|^2 = \pi/(|t| \sinh(\pi|t|)) \leq 7/\cosh(\pi t)$.

The range $1 < t_j \leq T_1$. We apply Lemma 10.6 with $\xi_j(n) = \rho_j(n)$, $\omega_j = |\Gamma(\pm it_j)|^2$ and

$$\varphi_j(y) = \overline{\psi_0(y/M)(\pi yY)^{\mp it_j} M_y(\pm it_j)} \quad (y \in [M, 2M]),$$

where M_y is defined for real $y > 0$ by the same formula. By Lemma 11.3, $|\varphi_j| \ll 1$. Moreover

$$|\varphi'_j(y)| \ll \frac{1}{M} + \frac{t_j}{M} + \frac{X}{K} \ll \frac{T_1}{M},$$

because $X/K \leq 2(qX)^\eta/M$ by $M \leq 2h_0$. Lemma 10.6 and Lemma 10.2 (with $a_n = \mathbf{1}\{M < n \leq y\}$) give

$$\begin{aligned} \sum_{1 < t_j \leq T_1} |B_j^\pm|^2 &\ll \frac{X^2}{K^2} Y^{-1} T_1^2 \sup_{M \leq y \leq 2M} \sum_{t_j \in [0, T_1]} \frac{1}{\cosh(\pi t_j)} \left| \sum_{M < n \leq y} \rho_j(n) \right|^2 \\ &\ll \frac{X^2}{K^2} Y^{-1} T_1^2 \left(T_1^2 + \frac{M^{1+\eta}}{q} \right) M. \end{aligned}$$

With $\sum_{t_j \leq T_1} |U_j|^2 \ll T_1^2 N^{1+\eta}$ (Lemma 10.7) and Cauchy–Schwarz, the contribution of these terms to Ξ_M is

$$\ll T_1^3 \frac{X}{K} Y^{-1/2} N^{1/2+\eta} \left(\left(1 + \frac{M^{1+\eta}}{q} \right) M \right)^{1/2}.$$

Since $T_1^3 \leq L^3(qX)^{3\eta}$, this is bounded by the right-hand side of (12.2).

The range $t_j > T_1$. Let $k'_\eta = \lceil 20/\eta \rceil$. By Lemma 11.3, $|M_h(\pm it_j)| \ll L^{k'_\eta} t_j^{-k'_\eta}$, so $|B_j^\pm| \ll (X/K)Y^{-1/2} |\Gamma(it_j)| L^{k'_\eta} t_j^{-k'_\eta} \sum_{h \sim M} |\rho_j(h)|$. For $T = 2^k T_1$, $k \geq 0$, Cauchy–Schwarz over h and Lemma 10.2 give

$$\begin{aligned} \sum_{T < t_j \leq 2T} |B_j^\pm|^2 &\ll \frac{X^2}{K^2} Y^{-1} L^{2k'_\eta} T^{-2k'_\eta} M \sum_{h \sim M} \sum_{t_j \leq 2T} \frac{|\rho_j(h)|^2}{\cosh(\pi t_j)} \\ &\ll \frac{X^2}{K^2} Y^{-1} L^{2k'_\eta} T^{2-2k'_\eta} M^3, \end{aligned}$$

where we used $M^{1+\eta}/q \leq MT^2$. Together with $\sum_{t_j \leq 2T} |U_j|^2 \ll T^2 N^{1+\eta}$, the contribution of the block is

$$\ll \frac{X}{K} Y^{-1/2} N^{1/2+\eta} L^{k'_\eta} T^{2-k'_\eta} M^{3/2}.$$

Summing over k gives a factor $\ll L^{k'_\eta} T_1^{2-k'_\eta} = L^2(qX)^{-\eta(k'_\eta-2)} \leq L^2(qX)^{-18}$. By the polynomial bounds, the total is $\ll L^2(qX)^{-4}$.

12.4. **The spectrum $\mathcal{J}_1$: the remainder.** We have

$$B_j^R = \frac{X}{K} Y^{-1/2} \frac{1}{2\pi i} \int_{(-3/2)} G_{t_j}(w) D_j(w) dw, \quad D_j(w) = \sum_h \overline{\rho_j(h)} b_h(w),$$

where

$$b_h(w) = \psi_0\left(\frac{h}{M}\right) (2\pi h Y)^{-w} M_h(w).$$

The coefficients $b_h(w)$ do not depend on j . Write $w = -\frac{3}{2} + i\xi$. By Lemma 11.3 with $k = 6$ and by (12.1),

$$|b_h(w)| \ll (MY)^{3/2} L^6 (1 + |\xi|)^{-6}, \quad \text{so} \quad \sum_h |b_h(w)|^2 \ll M(MY)^3 L^{12} (1 + |\xi|)^{-12}.$$

By Cauchy–Schwarz with respect to the measure $(1 + |\xi|)^{-2} d\xi$, which has total mass 2,

$$|B_j^R|^2 \leq \frac{X^2}{K^2} Y^{-1} \frac{2}{4\pi^2} \int_{\mathbb{R}} (1 + |\xi|)^2 |G_{t_j}(w)|^2 |D_j(w)|^2 d\xi.$$

Let $T = 2^k$ with $k \geq 0$, and let $T < t_j \leq 2T$. If $|\xi| \leq T/2$, then $|\xi \pm t_j| \geq T/2$, so Lemma 11.2 gives $|G_{t_j}(w)|^2 \ll e^{-\pi t_j} (1 + T)^{-5}$. For all ξ it gives $|G_{t_j}(w)|^2 \ll e^{-\pi t_j}$. Since $e^{-\pi t} \leq 1/\cosh(\pi t)$, Lemma 10.2 gives, for each ξ ,

$$\sum_{T < t_j \leq 2T} e^{-\pi t_j} |D_j(w)|^2 \ll \left(T^2 + \frac{M^{1+\eta}}{q}\right) M(MY)^3 L^{12} (1 + |\xi|)^{-12}.$$

Therefore

$$\sum_{T < t_j \leq 2T} |B_j^R|^2 \ll \frac{X^2}{K^2} Y^{-1} \left(T^2 + \frac{M^{1+\eta}}{q}\right) M(MY)^3 L^{12} \int_{\mathbb{R}} (1 + |\xi|)^{-10} ((1 + T)^{-5} + \mathbf{1}\{|\xi| > T/2\}) d\xi,$$

and the integral is $\ll (1 + T)^{-5}$. With $\sum_{t_j \leq 2T} |U_j|^2 \ll T^2 N^{1+\eta}$, the contribution of the block $T < t_j \leq 2T$ is

$$\ll \frac{X}{K} Y^{-1/2} N^{1/2+\eta} L^6 (MY)^{3/2} M^{1/2} \left(T^{-1/2} + \left(\frac{M^{1+\eta}}{q}\right)^{1/2} T^{-3/2}\right).$$

The sum over $k \geq 0$ converges, and $(MY)^{3/2} \leq (qX)^{3\eta/2}$ by (12.1). So the total contribution of the remainder terms is bounded by the right-hand side of (12.2). It is here that we need the decay of G_t in t given by Lemma 11.2: for the block $T < t_j \leq 2T$, the Heegner side and the large sieve each contribute a factor T^2 , and the factor $(1 + T)^{-5}$ compensates for both.

12.5. **The continuous spectrum.** The Eisenstein terms are treated in the same way. The sum over j is replaced by $\frac{1}{4\pi} \sum_{\mathfrak{s}} \int dr$, t_j by r , $\rho_j(h)$ by $\rho_{\mathfrak{s}}(h, r)$ and U_j by $U_{\mathfrak{s}}(r)$, and the Cauchy–Schwarz inequality is applied to $\frac{1}{4\pi} \sum_{\mathfrak{s}} \int dr$ in place of $\sum_j$. For $|r| \leq 1$ we use Lemma 11.1, Cauchy–Schwarz in σ , and the Eisenstein part of Lemma 10.2 with $T = 1$, as for real $t_j \in [0, 1]$. For $1 < |r| \leq T_1$ we use Lemma 11.2, the bound $|\Gamma(ir)|^2 \leq 7/\cosh(\pi r)$, and Lemma 10.6; the proof of Lemma 10.6 applies verbatim to a family indexed by $(\mathfrak{s}, r)$ with the measure $\frac{1}{4\pi} dr$ in place of a countable family. For $|r| > T_1$, and for the remainder terms, we split the range of $|r|$ into the intervals $(T, 2T]$ as in §§12.3–12.4. On the Heegner side we use the Eisenstein part of Lemma 10.7. There is no exceptional contribution, because r is real.

12.6. Completion of the proof. Collecting the estimates, we obtain (12.2). Since $Y^{-1/2} = (EK)^{1/2}N^{-1/4}$, we have

$$\frac{X}{K}Y^{-1/2}N^{1/2} = E^{1/2}N^{1/4}XK^{-1/2}.$$

Lemma 12.1. For $\frac{1}{2} \leq M \leq 2h_0$,

$$XK^{-1/2} \left(\left(1 + \frac{M}{q}\right)M \right)^{1/2} \mathfrak{E}_P \leq 6(qX)^\eta X^{1/2} \left(1 + \frac{X}{dH^{1/2}}\right)^\theta$$

and

$$XK^{-1/2} \left(\left(1 + \frac{M}{q}\right)M \right)^{1/2} \mathfrak{E}_{DI} \leq 6(qX)^{2\eta} X^{1/2} \left(1 + \frac{K}{dH^{1/2}}\right)^\theta.$$

Proof. Recall that $V = (qX)^\eta EK/(M\sqrt{N})$, $2h_0 = 2(qX)^\eta K/X$, and $(1+\alpha)^\theta \leq 1+\alpha^\theta \leq 2(1+\alpha)^\theta$ for $\alpha \geq 0$. We also use $EX/(q\sqrt{N}) = X/(4d\sqrt{N}) \leq X/(dH^{1/2})$.

First bound, $M \leq q$. Then $(1 + M/q)M \leq 2M$, and $V/\max(M, q) = \alpha_1/M$ with

$$\alpha_1 = \frac{(qX)^\eta EK}{q\sqrt{N}}.$$

The left-hand side is at most

$$\sqrt{2}XK^{-1/2}(M^{1/2} + \alpha_1^\theta M^{1/2-\theta}) \leq \sqrt{2}XK^{-1/2}(2h_0)^{1/2} \left(1 + \left(\frac{\alpha_1}{2h_0}\right)^\theta\right),$$

since $\theta \leq \frac{1}{2}$. Here $XK^{-1/2}(2h_0)^{1/2} = \sqrt{2}(qX)^{\eta/2}X^{1/2}$ and $\alpha_1/(2h_0) = EX/(2q\sqrt{N})$.

First bound, $q < M \leq 2h_0$. Then $(1 + M/q)M \leq 2M^2/q$ and $V/\max(M, q) = \beta_1/M^2$ with $\beta_1 = (qX)^\eta EK/\sqrt{N}$. The left-hand side is at most

$$\frac{\sqrt{2}X}{(Kq)^{1/2}}(M + \beta_1^\theta M^{1-2\theta}) \leq \frac{\sqrt{2}X}{(Kq)^{1/2}}(2h_0) \left(1 + \left(\frac{\beta_1}{(2h_0)^2}\right)^\theta\right).$$

Put $\varsigma = K/(qX) \leq 1$. Then $XK^{-1/2}q^{-1/2}(2h_0) = 2(qX)^\eta X^{1/2}\varsigma^{1/2}$, and

$$\frac{\beta_1}{(2h_0)^2} \leq \frac{EX^2}{4K\sqrt{N}} = \frac{\gamma_1}{\varsigma}, \quad \gamma_1 = \frac{EX}{4q\sqrt{N}}.$$

Since $\varsigma^{1/2} + \varsigma^{1/2-\theta}\gamma_1^\theta \leq 1 + \gamma_1^\theta$, the first bound follows in both cases.

Second bound. If $M \leq q$ then $V/\max(1, q/M) = VM/q$, and if $M > q$ then $V/\max(1, q/M) = V < VM/q$. In both cases $V/\max(1, q/M) \leq (qX)^\eta EK/(q\sqrt{N}) \leq (qX)^\eta K/(dH^{1/2})$, so $\mathfrak{E}_{DI} \leq (qX)^\eta(1 + K/(dH^{1/2}))^\theta$. Moreover

$$XK^{-1/2} \left(\left(1 + \frac{M}{q}\right)M \right)^{1/2} \leq XK^{-1/2} \left((2h_0)^{1/2} + \frac{2h_0}{q^{1/2}} \right) \leq \sqrt{2}(qX)^{\eta/2}X^{1/2} + 2(qX)^\eta \left(\frac{K}{q}\right)^{1/2},$$

and $(K/q)^{1/2} \leq X^{1/2}$. □

By (12.2) and Lemma 12.1,

$$\Xi_M(\psi_2) \ll L^{O(1)}(qXN)^{O(\eta)}N^{1/4}X^{1/2}(1 + \mathcal{A})^\theta + L^2(qX)^{-4},$$

with $\mathcal{A} = X/(dH^{1/2})$, respectively $\mathcal{A} = K/(dH^{1/2})$. By Lemma 9.1(d) we sum this over the $O(\log qX)$ values $M \in \mathcal{M}$, for ψ_2 and for ψ_2^- , and add the error term $O(\Delta^{k\eta}(qX)^{-1})$. Since $L \leq 9\Delta(qX)^\eta$, $N \leq 8H$ and $N \leq X^2$, renaming η proves Theorem 6.1. □

Remark 12.2. For general sequences of length M , [23, Theorem A] allows the range

$$\mathcal{Y} \ll \max(1, q/M, q^2/M^3),$$

which is larger than $\max(1, q/M)$ when $M < q^{1/2}$. Using it improves the contribution of the frequencies $M < q^{1/2}$ in the proof of Theorem 6.1(b), but it does not improve Theorem 1.3. In the application the largest moduli are dk_+ , and $(dk_+)^2 \asymp qX^2$ by (7.2). So the decisive frequencies have $M \asymp dk_+/X \asymp q^{1/2}$, up to factors $(qX)^{O(\eta)}$, and there the two ranges agree.

Remark 12.3. If $\theta = 0$ is admissible, there is no exceptional spectrum. Then Lemmas 10.3, 10.4 and 10.5 are not needed, and Theorem 6.1 reads $\Xi \ll \Delta^{C(\eta)}(qXH)^\eta X^{1/2} H^{1/4}$.

13. REMARKS

Remark 13.1 (Where the loss is). The exponent of Theorem 1.1 comes from comparing the main term $x^{1/2}d^{-1/2}$ in Proposition 7.1 with the error term $x^{1/4+\theta/2}d^{3/4-3\theta/2}$. Apart from the exceptional factor, the error term comes from the bound $X^{1/2}H^{1/4}$ of Theorem 6.1. Its proof bounds $\sum_j B_j U_j$ by the Cauchy–Schwarz inequality, so we compare each of the two factors with its diagonal size, and then the resulting bound with the observed size of Ξ .

The frequency side. The bounds for $\sum_j |B_j|^2$ in Section 12 are $(X/K)^2 Y^{-1} M$, which is the diagonal term of the large sieve, times the factor $1 + M/q \leq 3(qX)^\eta$ coming from the Kloosterman term, and factors $(qX)^{O(\eta)}$. Large spectral parameters contribute little: the main terms of the Bessel transform are negligible for $|t| > T_1$, because $2\pi h Y s \ll (qX)^\eta$, and the remainder terms decay in $|t|$.

The Heegner side. Here the bound $\mathcal{P} \ll N^{1+\eta}$ of Lemma 8.4 is far from the diagonal size. The quantity $\mathcal{P}$ counts the pairs (i, i') and the $\gamma \in \Gamma_0(q)$ with $u(z_i, \gamma z_{i'}) \leq 1$. In the proof of Lemma 8.4 it is bounded after enlarging $\Gamma_0(q)$ to $\mathrm{SL}_2(\mathbb{Z})$, by counting pairs of Heegner points of discriminant $-4N$ at bounded distance on $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$. For that count the bound is essentially sharp. Since a disc $\{w : u(w, z) \leq 1\}$ has hyperbolic area 4π and a fundamental domain for $\mathrm{SL}_2(\mathbb{Z})$ has area $\pi/3$, the equidistribution of Heegner points [5] suggests that there are about $12(\#\Lambda_N)^2 = N^{1+o(1)}$ pairs $(\mathfrak{z}, \mathfrak{w}) \in \Lambda_N \times \mathcal{Q}_N$ with $u(z(\mathfrak{w}), z(\mathfrak{z})) \leq 1$, and the computations in Section 14 confirm this. The quantity $\mathcal{P}$ itself is much smaller. In the proof of Lemma 8.4 we dropped the condition $\tau' \diamond \mathfrak{z}' \in \mathcal{Q}^\kappa$, which is equivalent to $\tau_i \diamond \mathfrak{w} \in \mathcal{Q}^\kappa$ for $\mathfrak{w} = g \diamond \mathfrak{z}'$. Together with $\tau_i \diamond \mathfrak{z}_i \in \mathcal{Q}^\kappa$, it implies that the binary forms of $\mathfrak{z}_i$ and $\mathfrak{w}$ have a common zero in $\mathbb{P}^1(\mathbb{F}_d)$, namely the image of the bottom row of τ_i , and this happens for only about $4/d$ of the pairs (Section 14). If the $I \leq N^{1/2+o(1)}$ points z_i were spread out randomly on $\Gamma_0(q) \backslash \mathbb{H}$, which has area $\asymp q$, we would have

$$\mathcal{P} \approx 2I + \frac{I^2}{q} \approx N^{1/2+o(1)} + \frac{N^{1+o(1)}}{d}.$$

In the examples of Section 14 the exact value of $\mathcal{P}$ is at most twice its diagonal term $2I$, which is consistent with this heuristic. Equivalently, the heuristic says that $U_j = \sum_i u_j(z_i)$ behaves like a sum of I random terms of size $q^{-1/2}$ for the $\asymp q$ forms with $|t_j| \leq 1$. Bounds of this strength would amount to a pair-correlation statement for Heegner points of discriminant $-4N$ on $X_0(d)$ at the scale of their mean spacing, which seems out of reach; compare [5] for equidistribution at a fixed scale. Sums of u_j over Heegner points, twisted by characters of the class group, are related by Waldspurger’s formula [28] to central values of Rankin–Selberg L -functions of u_j and theta series, but this family is too small compared with the conductors for current moment methods. Where the level varies and the polynomial is fixed, as in [13], the analogous congruence can be exploited by summing over the level; at a single level we do not see how to exploit it.

Since $N \asymp d^2$ in the application, a bound $\mathcal{P} \ll N^{1/2+\eta} + N^{1+\eta}/d$ would save a factor $N^{1/4} \asymp d^{1/2}$ in Theorem 6.1. The error term $x^{1/4+\varepsilon} d^{1/2+\varepsilon}$ in Lemma 7.2 can be improved as well: shifting

the contour in the Mellin integral representing $G(y)$ to $\operatorname{Re} w = -\frac{1}{2} + \varepsilon$ and using the convexity bound for $L(s, \psi)$ gives $\mathfrak{L}_d - G(y) \ll y^{-1/2+\varepsilon} d^{1/2+\varepsilon}$. With both improvements, and with the range $d \leq x^{1/4}$ of (7.1) relaxed to $d \leq x^{1/3-\varepsilon}$, Theorem 6.1(a) would give every $\delta < 1/6$, for every $\theta < \frac{1}{2}$. With Theorem 6.1(b) it would give every $\delta < (1 - 2\theta)/(6 - 8\theta)$, which is $25/164$ for $\theta = 7/64$.

The Cauchy–Schwarz step. Even with the diagonal sizes of both factors, the method stops short of the truth. With these sizes the Cauchy–Schwarz inequality gives $\Xi \ll X^{1/2+o(1)}$, apart from the exceptional factor. For $K \geq 4X$, however, the Type I sums computed in Section 14 are at most the square root of the main term $\mathfrak{M} = X^{1+o(1)}/d$ of Ξ , that is, of size about $(X/d)^{1/2}$. At $K = X$, where they are largest among the values computed, they are smaller than $X^{1/2}$ by factors between 7 and 43, for d between 53 and 281. The Cauchy–Schwarz inequality cannot detect cancellation among the $\asymp q$ spectral terms with $|t_j| \leq 1$: in the random model, $\sum_j B_j U_j$ is about $q^{-1/2}(\sum_j |B_j|^2)^{1/2}(\sum_j |U_j|^2)^{1/2}$. So, relative to the observed size, the method loses a further factor of about $q^{1/2} \asymp d^{1/2}$ in this step.

This loss does not matter for exponents up to $1/6$, which appears to be the limit for arguments that treat the primes d separately. For $d > x^{1/3}$ the polynomial T_d takes fewer than d values up to x , fewer than the number $h(-4N_d) = N_d^{1/2+o(1)}$ of classes of forms that govern its roots. Going beyond requires averaging over d . For the patterns with $B = 1$, such as $\{0, 1, 2\}$, the relation $2dQ = u^2 + d^2 + 1$ says that $\begin{pmatrix} d & u \\ u & 2Q-d \end{pmatrix} = \sigma^t \sigma$ with $\sigma \in \operatorname{SL}_2(\mathbb{Z})$, so that $2Q = \|\sigma\|^2$, and averaging over d becomes a question about the distribution, in residue classes, of norms of hyperbolic lattice points; compare [10].

Remark 13.2 (The exceptional spectrum). With only the inequalities of [4], the exceptional factor in Theorem 6.1 is $(1 + K/(dH^{1/2}))^\theta$ instead of $(1 + X/(dH^{1/2}))^\theta$. In the application K/X reaches about $d^{1/2}$, so the error term grows by a factor $d^{\theta/2}$, and δ drops from $(1 - 2\theta)/(10 - 12\theta)$ to $(1 - 2\theta)/(10 - 8\theta)$; for $\theta = 7/64$ these are 0.0899 and 0.0856, and with $\theta = 1/4$ the inequalities of [4] alone give $\delta < 1/16$. The point of [23, Theorem 2] is that the sequences $\psi_0(n/M) \overline{c_n(\sigma)}$ that occur in §12.2 are smooth, and for such sequences the exceptional weight $V^{2\theta_j}$ is admissible in the range $V \leq \max(M, q)$ instead of $V \leq \max(1, q/M)$. At the decisive frequencies $M \asymp K/X$ one has $V \asymp EX/\sqrt{N}$, and the factor $(1 + V/q)^\theta$ becomes $(1 + X/(dH^{1/2}))^\theta$, up to factors $(qX)^{O(\eta)}$.

Remark 13.3 (Why the hyperbola method is kept). One might try to evaluate $\Sigma_d(x)$ without Lemma 5.1, by applying Theorem 6.1 to all divisors of $T' = (E\ell^2 + H)/d$. The moduli would then go up to about X^2 , far beyond the range $K \leq qX$ of Theorem 6.1, and in that range no estimate of this kind can hold. Indeed, the moduli $\mu = E\ell^2 + H$, with complementary divisor 1, all satisfy $\mu \equiv H \pmod{4}$, so they lie in only one of the two classes κ . For $K \asymp X^2$ about X/d of them lie in $[K, 2K]$ and are divisible by d , which is the size of the expected main term. The hyperbola identity avoids such moduli. It needs T to be 2^{v-2} times a number $\equiv 1 \pmod{4}$ for all ℓ in a lattice, and this is what forces the 2-adic analysis of Section 3.

Remark 13.4 (Other treatments of the Type I sums). The Weyl sums $S(h; \mu)$ can also be treated by the method of Hooley [14], which is the analytic engine of the work of Friedlander and Iwaniec on quadratic polynomials and quadratic forms [9]. The roots of $E\nu^2 + H \equiv 0 \pmod{\mu}$ are parametrised by binary quadratic forms, as in Lemma 8.2, the sums $S(h; \mu)$ become incomplete Kloosterman sums, and these are estimated by completion and Weil’s bound. In an earlier version of this work (§1.3) this was carried out with polynomial dependence on d . For moduli of size K it saves a factor $K^{1/4}$ over the trivial bound, at the cost of a power of d , and in our application it admits primes $d \leq x^{1/12-\varepsilon}$, which gives $\delta < 1/24$.

Duke, Friedlander and Iwaniec [7, Theorem 1.1] bound the sums

$$\sum_{c \equiv 0 \pmod{q}} f(c) \sum_{\substack{b \pmod{c} \\ b^2 \equiv D \pmod{c}}} e\left(\frac{hb}{c}\right),$$

for D a positive odd fundamental discriminant and f a suitably normalised smooth function supported on $[Y, 2Y]$, by $h^{1/4}(Y + h\sqrt{D})^{3/4}D^{1/8-1/1331}$, uniformly in D and q . They note that the condition that D be odd can be removed and that non-fundamental discriminants can also be treated, and in [7, §16] they state the same bound for negative fundamental discriminants. Its strength is the uniformity in D , which makes it non-trivial even for Y somewhat smaller than $\sqrt{D}$. In our application the moduli are odd, and the relevant discriminant is $-EH_d$, that is, $-4H_d$ or $-8H_d$. It has size d^2 , and since $H_d \equiv 1 \pmod{4}$ by Proposition 3.2, it is fundamental exactly when H_d is squarefree. The moduli go up to $d\sqrt{x}$, far beyond $\sqrt{D}$. In this range the bound saves the same power $Y^{1/4}$ as the estimate obtained from Weil's bound, so where it applies it would not improve the exponent $1/24$ significantly. The method of [6, 7], in which the roots are parametrised by quadratic forms and the resulting Weyl sums are analysed spectrally, is the starting point of the present paper.

Remark 13.5 (Automorphic kernels). In an earlier version (§1.3), a bound of the same strength as Theorem 6.1(a) was deduced from a bound of Grimmelt and Merikoski for weighted averages of automorphic kernels [12, Theorem 8.1], which they developed for the greatest prime factor of quadratic polynomials [13]. We sketch the connection; it is not used in this paper. By Lemmas 8.1 and 8.2 and (8.1), for any function f on $\mathbb{H}$ with compact support,

$$\sum_{\mathfrak{g} \in \mathcal{Q}^\kappa} f(z(\mathfrak{g})) = \frac{1}{2} \sum_{i=1}^I \sum_{\gamma \in \Gamma} f(\gamma z_i).$$

With $f(z) = \psi_1(\sqrt{N}/(EK \operatorname{Im} z))\psi_2(\sqrt{N} \operatorname{Re} z/(EX \operatorname{Im} z))$ the left-hand side is the double sum $\sum_\mu \psi_1(\mu/K) \sum_\ell \psi_2(\ell/X)$ in (6.1), so this sum is an average of an automorphic kernel over the Heegner points. Theorem 8.1 of [12] bounds such averages, minus their expected values, by the product of $(EX/\sqrt{N})^{1/2+o(1)}$, a factor accounting for the exceptional spectrum, and the square roots of two quantities of the same shape built from explicit non-negative kernels. For the Heegner points the relevant quantity is bounded by the count $\mathcal{P}$ of Lemma 8.4, and for the point i by an elementary count of matrices in $\Gamma_0(q)$. The ingredients of this route correspond closely to those of the proof given here: the spectral expansion of the kernel to Lemma 10.1; the two kernel quantities to the Heegner side and to the frequency side, where the matrices with lower left entry zero and non-zero correspond to the diagonal and Kloosterman terms of Lemma 10.2; and the condition on the parameters of [12, Theorem 8.1] to the condition that the exceptional weight $V^{2\theta_j}$ lies in the range of Lemma 10.4. Both routes give the same exceptional factor $(1 + X/(dH^{1/2}))^\theta$. The correspondence is heuristic; we do not claim an implication between [12, Theorem 8.1] and [23, Theorem 2]. The kernel route is shorter once [12, Theorem 8.1] is available, since it needs no Bessel functions. The route taken here rests only on published inputs, and it separates the sources of loss: improving the Heegner side would improve both routes equally, while the bound for the frequency side is already of the size of its diagonal term.

Remark 13.6 (Effectivity). Siegel's theorem enters only through Lemma 5.3(d), and it can be avoided. Let s be the squarefree part of m_d . By the Landau–Page theorem, at most one primitive real character ψ' of conductor $\leq x$ has $L(1, \psi') < c_{\text{LP}}/\log x$, for a suitable absolute constant $c_{\text{LP}} > 0$. The primitive character attached to d depends only on s , and different s give different primitive characters. For each s , the equation $(d+c)^2 - sy^2 = -4A(B-A)$ has $O(\log x)$ solutions with $0 < d \leq x$, uniformly in s : by the classical correspondence between primitive representations

by a binary quadratic form and roots of quadratic congruences, its solutions form $O_{A,B}(1)$ orbits under the group of units of norm 1 in $\mathbb{Z}[\sqrt{s}]$, and every such unit $\epsilon > 1$ satisfies $\epsilon \geq \sqrt{2} + \sqrt{3} > 3$. Discarding these $O(\log x)$ primes d leaves $\mathfrak{L}_d \gg (\log x)^{-1}(\log \log x)^{-2}$ effectively, for $d \leq x^{1/4}$. The other inputs, in particular the large sieve inequalities and the admissibility of θ , are effective.

Remark 13.7 (Correlations of r). Let n arise from some (d, ℓ) in (4.3). By the proof of Lemma 2.5, no prime $\ell' \equiv 3 \pmod{4}$ divides both P and $P + B$. Since $r/4$ is multiplicative, it follows that $r(P(P+B)) \leq \frac{1}{4}r(P)r(P+B)$. Combined with Lemma 2.6 and the proof of Proposition 4.2, this gives

$$\sum_{d \text{ good}} \Sigma_d(x) \leq \frac{1}{4} \sum_{n \leq x} r(n) r(n+a) r(n+b).$$

Hence $\sum_{n \leq x} r(n)r(n+a)r(n+b) \gg x^{1/2+\delta}$ for every $\delta < \delta_\theta$. Conjecturally this correlation is $\asymp x$. For the divisor function, by contrast, the triple correlation $\sum_{n \leq x} \tau(n-j)\tau(n)\tau(n+j)$ is known to have the conjectured order of magnitude $x(\log x)^3$; see [21].

14. NUMERICAL CHECKS

The scripts are in the folder `numerics` that accompanies this paper; see its `README.md`. They use Python 3 with `numpy` and `scipy`, and one C program.

The 2-adic set-up. For each of the 3725 patterns $\{0, a, b\}$ with $b \leq 100$ and a or b odd, we chose (p_0, B, A) by Table 2, constructed n_* following the proof of Lemma 3.1 (in its last subcase the class of o is found by a short search), and computed t_* , d_* , v and d_0 as in the proof of Proposition 3.2. We then checked the conclusion of Proposition 3.2 for 40 random pairs (d, u) with $d \equiv d_0 \pmod{2^J}$ and $u \equiv 0 \pmod{2^j}$ for each pattern. There were no failures. Table 3 lists some of the data.

pattern	p_0	B	A	n_*	v	J	d_0	E	m_d
$\{0, 1, 2\}$	0	1	2	16	3	5	25	8	$d^2 + 6d + 1$
$\{0, 1, 3\}$	0	3	1	1	3	5	1	8	$d^2 - 2d + 9$
$\{0, 2, 3\}$	2	1	-2	20	3	5	21	8	$d^2 - 10d + 1$
$\{0, 1, 4\}$	0	1	4	16	4	6	25	4	$d^2 + 14d + 1$
$\{0, 3, 8\}$	0	3	8	77	2	4	5	4	$d^2 + 26d + 9$
$\{0, 5, 8\}$	0	5	8	320	5	7	53	8	$d^2 + 22d + 25$

TABLE 3. The data of Lemmas 2.3 and 3.1 and Proposition 3.2 for some patterns.

The main term. Table 4 compares the sums $\Sigma_d^+(x)$, defined as in (4.3) but over $\ell \geq 1$ only, with half of the main term of Proposition 7.1, at $x = 10^{13}$. For each pattern we took the first good prime $d > d_1$ and the first good prime $d > 1000$. We used $F(t) = \exp(-1/(20(t-\frac{1}{2})(1-t)))$ for $\frac{1}{2} < t < 1$, for which $c_F = 0.16684$. The constant $\mathfrak{L}_d$ was evaluated as $L(1, \chi)L(1, \psi)\mathcal{P}_d$ (Lemma 5.3(c)), with $L(1, \psi)$ computed from the rapidly convergent series supplied by the functional equation. The last column gives the corresponding ratio for the sharp cut-off, that is, for $\sum_{\ell \geq 1} r(T_d(\ell))$ over $T_d(\ell) \leq x$, compared with the same formula with c_F replaced by 2; this case is not covered by Proposition 7.1. In all these computations every $T_d(\ell)$ was an integer of the form $2^{v-2}T'$ with $T' \equiv 1 \pmod{4}$, as Proposition 3.2 predicts, and $P, P+B \in \mathcal{S}$ in every case, as Lemma 2.5 predicts.

Type I sums. We computed the sums Ξ of (6.1) for $(E, H) = (8, 1057)$ and $(8, 10081)$, which come from the pattern $\{0, 1, 2\}$ with $d = 89$ and $d = 281$, and for $(E, H) = (8, 285)$ and $(8, 3869)$, which come from $\{0, 2, 3\}$ with $d = 53$ and $d = 181$. We took $\psi_1(t) = \phi(2t - 3)$ and $\psi_2 = \phi$

pattern	d	$\#\{\ell\}$	$\Sigma_d^+(x)$	main term/2	ratio	sharp ratio
$\{0, 1, 2\}$	89	167 600	51 778	51 150	1.012	1.004
	1049	48 818	13 851	13 544	1.023	1.013
$\{0, 1, 3\}$	97	160 540	47 194	47 438	0.995	0.997
	1153	46 564	8 362	8 280	1.010	1.012
$\{0, 2, 3\}$	53	217 186	90 690	90 209	1.005	1.002
	1013	49 678	28 238	28 206	1.001	1.004
$\{0, 1, 4\}$	89	167 600	58 831	58 606	1.004	1.001
	1049	48 818	16 275	16 162	1.007	1.008
$\{0, 5, 8\}$	181	58 763	22 534	22 796	0.989	1.002
	1973	17 798	12 985	13 142	0.988	0.992

TABLE 4. The sums $\Sigma_d^+(x)$ at $x = 10^{13}$ compared with half of the main term $32c_F \mathfrak{L}_d 2^{-j} \sqrt{x/d}$ of Proposition 7.1. Here $\#\{\ell\}$ is the number of $\ell \geq 1$ with $d \mid E\ell^2 + H_d$ and $T_d(\ell) \leq x$.

with $\phi(t) = \exp(-1/(1-t^2))$, $X = 2 \cdot 10^6$, $\kappa \in \{1, 3\}$, and $K = 2^{i/2}X$ for $0 \leq i \leq 2 \log_2 d$. In all cases $|\Xi| \leq 0.03 X^{1/2} H^{1/4}$; the maxima of $|\Xi|/(X^{1/2} H^{1/4})$ were 0.024, 0.003, 0.028 and 0.003. The largest values of $|\Xi|$ occur for K close to X , where $|\Xi|$ is at most 190, 48, 164 and 33, against $X^{1/2} = 1414$. For $K \geq 4X$ we always found $|\Xi|$ at most the square root of the main term $\mathfrak{M}$, which is roughly the number of terms. This suggests that the loss $H^{1/4}$ in Theorem 6.1 is a feature of the method (Remark 13.1).

Pairs of Heegner points. For $N = EH_d$ with the pattern $\{0, 1, 2\}$ and $d \in \{89, 281, 409\}$, and with the pattern $\{0, 2, 3\}$ and $d \in \{53, 181, 373\}$, we counted the pairs $(\mathfrak{z}, \mathfrak{w}) \in \Lambda_N \times \mathcal{Q}_N$ with $u(z(\mathfrak{w}), z(\mathfrak{z})) \leq 1$, which appear in Lemma 8.4. The number of pairs lay between $4.5N$ and $61N$, and in each case it agreed with $12(\#\Lambda_N)^2$ to within 1.5%, as equidistribution predicts (Remark 13.1). The number of pairs whose binary forms have a common zero in $\mathbb{P}^1(\mathbb{F}_d)$ was between $3.6/d$ and $4.1/d$ times the number of all pairs. Finally, for $(E, H, d) = (8, 1057, 89)$, $(8, 285, 53)$ and $(8, 3869, 181)$ and both values of κ , we computed I and the exact value of $\mathcal{P}$, by the reduction in the proof of Lemma 8.4 with the condition $\tau_i \diamond \mathfrak{w} \in \mathcal{Q}^\kappa$ kept. The values of $\mathcal{P}$ are 1664, 256 and 576, against the diagonal terms $2I = 896, 256$ and 576 and against $N = 8456, 2280$ and 30952 ; they do not depend on κ . At most 8 cosets τ satisfied $\tau \diamond \mathfrak{z} \in \mathcal{Q}^\kappa$, against the bound $12E = 96$ of Lemma 8.3.

The Heegner–Poincaré identity (Lemma 9.2). The proof of (9.2) does not use the smoothness of g . We checked (9.2) for the sharp weight $g = \mathbf{1}_{(\sqrt{N}/(2EK), \sqrt{N}/(EK)]}$, that is, for $\sum_{K \leq \mu < 2K} S(h; \mu)$. The data were $(E, H, d) = (8, 1057, 89)$ and $(8, 285, 53)$, $\kappa \in \{1, 3\}$, $K \in \{600, 2500, 20000, 60000\}$ and $h \in \{1, 2, 3, 7, 13\}$. Both sides were computed exactly, apart from the evaluation of the exponentials. The right-hand side runs over the $I = 448$, respectively 128, points z_i and over the cosets $\Gamma_\infty \gamma$ that contribute. The number of terms on each side was the same in all cases, up to 562, and the differences were below $1.3 \cdot 10^{-13}$.

Poisson summation and truncation (Lemma 9.1). For the same (E, H, d) and κ , with $X = 3000$, $K/X \in \{\frac{1}{2}, 1, 4, 16, 64\}$, $\psi_1(t) = \phi(2t - 3)$ and the non-even weight $\psi_2(t) = \phi(t)(1 + t/2)$, we computed Ξ directly and as $\sum_{h \neq 0} W_h$. In all 20 cases the two agree to within $4 \cdot 10^{-10}$. The partial sums over $0 < |h| \leq cK/X$ differ from Ξ by at most 0.11, 0.046, 0.0081 and 0.0012 for $c = 1, 2, 4, 8$. Here $|\Xi| \leq 0.51$ and $|\Xi| \leq 0.0023 X^{1/2} H^{1/4}$. We also confirmed $S(-h; \mu) = S(h; \mu)$ numerically.

The Bessel lemmas. For real t we computed $K_{it}(y)$ from the convergent series

$$K_{it}(y) = \operatorname{Re} \left[\Gamma(it) \left(\frac{y}{2} \right)^{-it} \sum_{k \geq 0} \frac{(-1)^k (y/2)^{2k}}{k! \prod_{m=1}^k (it - m)} \right],$$

which is the full residue expansion of the Mellin–Barnes integral in Lemma 11.2. Every term carries the factor $\Gamma(it)$, so there is no cancellation. The series agrees with a contour-shifted quadrature to relative accuracy $5 \cdot 10^{-14}$ for $t \leq 15$. The representation of Lemma 11.1 agrees with it, and with `scipy`'s K_ν for real ν , to relative accuracy $2.3 \cdot 10^{-10}$ for $\nu \in \{0, 0.3i, i, 7.5i, 0.05, 0.109375, 0.4\}$ and $y \in \{10^{-3}, 0.05, 0.7, 3\}$. The formula of Lemma 11.2, with the integral evaluated by the trapezoid rule, holds to within $4 \cdot 10^{-15} |t|^{-1/2} e^{-\pi|t|/2}$ for $|t| \in \{1, 2.5, 6, 15\}$ and $y \in \{10^{-3}, 0.02, 0.4, 2\}$. The terms $k \geq 1$ of the series give the remainder in Lemma 11.2 exactly. It is not pointwise $O(y^{3/2} |t|^{-5/2} e^{-\pi|t|/2})$: for small y it is of order $y^2 |t|^{-3/2} e^{-\pi|t|/2}$, and for $y = 2$ the maximum of its absolute value at the points ys , $s \in [\frac{1}{2}, 1]$, is $21.7 y^{3/2} |t|^{-5/2} e^{-\pi|t|/2}$ at $|t| = 25$. This is why the proof only uses its average against a smooth weight in s , through the factor $M_h(w)$. Such averages decay in $|t|$, as predicted. For example, for $y = 0.02$ the ratio of the average to $y^{3/2} e^{-\pi|t|/2}$ falls from $3.2 \cdot 10^{-3}$ at $|t| = 1$ to $1.2 \cdot 10^{-5}$ at $|t| = 15$ and $1.6 \cdot 10^{-6}$ at $|t| = 25$.

Square-root cancellation in the Weyl sums. For $(E, H, d) = (8, 1057, 89)$ and $(8, 285, 53)$, $\kappa \in \{1, 3\}$ and $K \in \{2 \cdot 10^4, 10^5, 3 \cdot 10^5\}$, let n be the number of pairs (μ, ν) with $K \leq \mu < 2K$, $\mu \equiv \kappa d \pmod{4d}$ and $E\nu^2 + H \equiv 0 \pmod{\mu}$, and let $S_h(K) = \sum_{K \leq \mu < 2K} S(h; \mu)$ over the same μ . Over $1 \leq h \leq 200$, the root mean square of $|S_h(K)|/\sqrt{n}$ lies between 0.82 and 0.99, and the maximum lies between 2.0 and 3.7, for n up to 2584.

Exponents. Table 1 was computed in exact arithmetic.

REFERENCES

- [1] T. Cochrane and R. E. Dressler, *Consecutive triples of sums of two squares*, Arch. Math. (Basel) **49** (1987), 301–304.
- [2] R. D. Connors and J. P. Keating, *Two-point spectral correlations for the square billiard*, J. Phys. A **30** (1997), 1817–1830.
- [3] H. Davenport, *Multiplicative Number Theory*, 3rd ed., Graduate Texts in Mathematics 74, Springer, 2000.
- [4] J.-M. Deshouillers and H. Iwaniec, *Kloosterman sums and Fourier coefficients of cusp forms*, Invent. Math. **70** (1982/83), 219–288.
- [5] W. Duke, *Hyperbolic distribution problems and half-integral weight Maass forms*, Invent. Math. **92** (1988), 73–90.
- [6] W. Duke, J. B. Friedlander and H. Iwaniec, *Equidistribution of roots of a quadratic congruence to prime moduli*, Ann. of Math. (2) **141** (1995), 423–441.
- [7] W. Duke, J. B. Friedlander and H. Iwaniec, *Weyl sums for quadratic roots*, Int. Math. Res. Not. IMRN **2012**, no. 11, 2493–2549.
- [8] T. Freiberg, P. Kurlberg and L. Rosenzweig, *Poisson distribution for gaps between sums of two squares and level spacings for toral point scatterers*, Commun. Number Theory Phys. **11** (2017), 837–877.
- [9] J. Friedlander and H. Iwaniec, *Quadratic polynomials and quadratic forms*, Acta Math. **141** (1978), 1–15.
- [10] J. Friedlander and H. Iwaniec, *Hyperbolic prime number theorem*, Acta Math. **202** (2009), 1–19.
- [11] I. S. Gradshteyn and I. M. Ryzhik, *Table of Integrals, Series, and Products*, 7th ed., Academic Press, 2007.
- [12] L. Grimmelt and J. Merikoski, *Weighted averages of $\mathrm{SL}_2(\mathbb{R})$ automorphic kernel, part I: non-oscillatory functions*, arXiv:2505.00489 (2025).
- [13] L. Grimmelt and J. Merikoski, *On the greatest prime factor and uniform equidistribution of quadratic polynomials*, arXiv:2505.00493 (2025).
- [14] C. Hooley, *On the number of divisors of quadratic polynomials*, Acta Math. **110** (1963), 97–114.
- [15] C. Hooley, *On the intervals between numbers that are sums of two squares. II*, J. Number Theory **5** (1973), 215–217.
- [16] C. Hooley, *On the intervals between numbers that are sums of two squares. III*, J. Reine Angew. Math. **267** (1974), 207–218.
- [17] K.-H. Indlekofer, *Scharfe untere Abschätzung für die Anzahlfunktion der B-Zwillinge*, Acta Arith. **26** (1974/75), 207–212.

- [18] H. Iwaniec, *Spectral Methods of Automorphic Forms*, 2nd ed., Graduate Studies in Mathematics 53, AMS, 2002.
- [19] H. H. Kim, *Functoriality for the exterior square of GL_4 and the symmetric fourth of GL_2* , with Appendix 1 by D. Ramakrishnan and Appendix 2 by H. H. Kim and P. Sarnak, J. Amer. Math. Soc. **16** (2003), 139–183.
- [20] N. V. Kuznetsov, *The Petersson conjecture for cusp forms of weight zero and the Linnik conjecture. Sums of Kloosterman sums*, Mat. Sb. (N.S.) **111(153)** (1980), 334–383; English transl. in Math. USSR-Sb. **39** (1981), 299–342.
- [21] B. Misra, M. Ram Murty and B. Saha, *A triple convolution sum of the divisor function*, arXiv:2508.13082 (2025).
- [22] W. G. Nowak, *On the distribution of M -tuples of B -numbers*, Publ. Inst. Math. (Beograd) (N.S.) **77(91)** (2005), 71–78.
- [23] A. Pascadi, *Large sieve inequalities for exceptional Maass forms and the greatest prime factor of $n^2 + 1$* , Forum Math. Pi **14** (2026), e8; arXiv:2404.04239.
- [24] A. Selberg, *On the estimation of Fourier coefficients of modular forms*, in: Theory of Numbers, Proc. Sympos. Pure Math. **8**, AMS, 1965, 1–15.
- [25] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, 1973.
- [26] M. Skalba, *A note on consecutive sums of two squares*, Bull. Pol. Acad. Sci. Math. **73** (2025), no. 2, 111–116.
- [27] Y. Smilansky, *Sums of two squares—pair correlation and distribution in short intervals*, Int. J. Number Theory **9** (2013), 1687–1711.
- [28] J.-L. Waldspurger, *Sur les valeurs de certaines fonctions L automorphes en leur centre de symétrie*, Compositio Math. **54** (1985), 173–242.
- [29] G. N. Watson, *A Treatise on the Theory of Bessel Functions*, 2nd ed., Cambridge University Press, 1944.